

The impact of green innovation network embeddedness on ESG performance in energy enterprises: The mediating role of green dynamic capabilities and the moderating role of organizational agility

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ABSTRACT

Under China's "dual carbon" strategy, energy enterprises increasingly rely on green innovation networks to improve their environmental, social, and governance (ESG) performance. Drawing on resource orchestration theory and dynamic capability theory, this study investigates how green innovation network structural embeddedness and green innovation network relational embeddedness influence ESG performance. It also examines the mediating role of green dynamic capability and the moderating role of organizational agility. Based on survey data collected from 566 energy enterprises in Northwestern China, the study applied hierarchical regression and nonlinear mediation analyses to uncover the underlying mechanisms. The results revealed that: (1) both green innovation network structural embeddedness and relational embeddedness exhibit an inverted U-shaped relationship with ESG performance; (2) green dynamic capabilities mediate the relationship between moderate network embeddedness and ESG performance, but the mediating effect becomes insignificant under conditions of high embeddedness; and (3) organizational agility positively moderates the impact of relational network embeddedness on both green dynamic capabilities and ESG performance, with moderated mediation effects observed, while no significant moderating effect is found in the structural embeddedness pathway. These findings offer theoretical insights and practical implications for energy enterprises seeking to enhance ESG performance through effective engagement in green innovation networks.

1. Introduction

Under China's "dual carbon" strategy and the accelerating global green transformation, enterprises face increasing pressure to enhance their environmental, social, and governance (ESG) performance as a key dimension of sustainable competitiveness (Jiang et al., 2023). Particularly in the energy sector, ESG performance is no longer merely a compliance issue but a strategic imperative that directly impacts firms' long-term viability and innovation capabilities. Energy enterprises, especially those in energy-intensive industries, are embedded in a web of green innovation networks comprised of government bodies, green technology providers, ESG service intermediaries, research institutions, and non-governmental organizations. These networks facilitate the exchange of knowledge, resources, and innovative solutions aimed at improving sustainability (Liu et al., 2021; Melander and Arvidsson, 2022). These networks enable firms to access advanced carbon

management tools, clean technologies and policy resources, helping them develop green governance capabilities.

However, the challenges of the green transition are particularly evident in Northwestern China, where industrialization pressures collide with green development mandates. This region, encompassing provinces like Shaanxi, Gansu and Inner Mongolia, is at the nexus of China's energy transformation, yet it faces substantial barriers, including insufficient green infrastructure, delayed transitions in organizational mindsets and lagging capability development (Xu, 2024). Many local energy enterprises, particularly those rooted in coal, petrochemical, and thermal power sectors, face complex constraints. They must not only maintain high industrial output but also comply with increasingly stringent ESG regulations. Due to these internal and external constraints, these firms often rely heavily on external green resources, such as green service providers and carbon management platforms, to upgrade their capabilities. These external dependencies give rise to green innovation

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network embeddedness, defined as the extent to which firms are structurally and relationally embedded in these green collaborative ecosystems (Granovetter, 1985; Peng et al., 2020).

While green innovation network embeddedness is widely regarded as a source of performance enhancement, recent studies suggest that excessive or poorly managed embeddedness can lead to diminishing returns or even negative outcomes (Zhang et al., 2023; Zhao et al., 2021). For example, energy firms may become over-embedded in a redundant set of green innovation platforms, leading to resource redundancy, ineffective knowledge integration and higher coordination costs. This phenomenon, sometimes referred to as the “green collaboration failure” or “green resource curse,” has been highlighted in recent empirical work by scholars such as Xie et al. (2024) and Wang et al. (2023). The Boston Consulting Group’s 2023 Global Green Transformation Report underscores these challenges, revealing that only 22 % of firms have experienced substantial performance benefits from green collaboration, while more than half report difficulties in coordination and integration.

Most existing literature emphasizes the positive effects of green innovation networks, yet few studies have explored the non-linear dynamics between network embeddedness and ESG performance (Teece, 2007; Wang et al., 2023). The traditional resource-based view (RBV) holds that firms achieve competitive advantage through the acquisition of valuable, rare, and inimitable resources (Barney, 2018), yet it fails to account for the unique characteristics of green resources—such as their high mobility, public availability, and institutional dependence—which make them inherently different from traditional strategic resources (Yu et al., 2024). In this context, green resources such as carbon technologies, ESG standards, and clean production processes exhibit higher generalizability and weaker path dependence than proprietary technologies, complicating the traditional RBV framework (Lampikoski et al., 2014).

To address these gaps, this study uses resource orchestration theory and dynamic capabilities theory to provide a comprehensive framework for understanding how green innovation network embeddedness influences ESG performance. Resource orchestration theory emphasizes the strategic management of resource portfolios, while dynamic capabilities theory focuses on firms’ ability to reconfigure and adapt their internal capabilities to align with external resource inflows (Salvato and Vassolo, 2017; Sirmon et al., 2011). These theoretical perspectives are essential for explaining how green innovation networks, characterized by heterogeneous resources and complex institutional environments, can enhance or undermine ESG performance. Recent studies, such as those by Wang and Zhang (2024), highlight the importance of resource orchestration in digital green supply chains and show how inter-organizational cooperation catalyzes eco-innovations through synchronized resource sharing and capability development. These insights reinforce the relevance of the dual theoretical approach employed in this study. This research also introduces the concept of organizational agility as a moderating factor that influences the relationship between green innovation network embeddedness and ESG performance. Organizational agility refers to a firm’s ability to respond quickly to environmental changes and reconfigure its internal processes (Puriwat and Hoonsopon, 2021). It is especially critical in the context of green innovation networks, where uncertainty and institutional complexity are prevalent (Rabal-Conesa et al., 2021).

This study develops and empirically tests a model in which green dynamic capabilities mediate the relationship between green innovation network embeddedness and ESG performance. Specifically, it explores the non-linear effects of structural and relational embeddedness, the mediating role of green dynamic capabilities, and the moderating effect of organizational agility on these relationships. Based on survey data from 566 energy enterprises in Northwestern China, this study aims to contribute to the literature by providing theoretical insights into the “blessing” and “curse” effects of green network embeddedness and offering practical recommendations for energy firms navigating the green

transition.

2. Research model and hypotheses development

2.1. Theoretical framework

The evolution of the resource-based view can be divided into three stages: the traditional resource-based view, the dynamic resource-based view, and the resource action perspective (Chen et al., 2021). The traditional resource-based view posits that a firm’s competitive advantage originates from its possession of VRIN resources—valuable, rare, inimitable, and non-substitutable assets (Helfat and Peteraf, 2003). However, this perspective focuses on a static stock of resources and fails to explain how firms dynamically acquire and transform resources in rapidly changing environments. To address this limitation, the theory of dynamic capabilities emerged, marking the second stage. This theory emphasizes a firm’s ability to build competitive advantage in uncertain environments through opportunity identification, resource integration, and reconfiguration (Salvato and Vassolo, 2017). Further, the resource action perspective aims to uncover the internal mechanisms linking resource acquisition, capability formation and performance realization (D’Oria et al., 2021). Represented by the theory of resource orchestration, it conceptualizes resource management as a systemic process consisting of three interrelated stages: structuring resource portfolios, bundling resources into capabilities, and leveraging capabilities to create value (Sirmon et al., 2011). Structuring refers to the accumulation of valuable resources and the removal of redundant ones to build a resource base aligned with strategic goals. Bundling involves transforming resources into specific capabilities that support performance outcomes. Leveraging represents the final step in which firms apply established capabilities to generate value.

The integration of resource orchestration theory and the dynamic capabilities perspective is particularly critical for understanding green innovation networks, given the unique attributes of green resources compared to traditional strategic resources. Unlike traditional VRIN resources—characterized by exclusivity, low mobility, and stable value—green resources are inherently more mobile, codified, and context-dependent (Yu et al., 2024). Their value is not realized through possession alone but requires continuous alignment with evolving policies, heterogeneous stakeholder expectations, and firm-specific operational contexts. This distinctiveness necessitates a dual theoretical lens: resource orchestration theory provides a macro-level process logic to explain how firms structure, bundle, and leverage green resources through networked collaboration, while the dynamic capabilities perspective offers a micro-foundational view of how firms sense, seize, and reconfigure these resources amid policy volatility and technological uncertainty.

Resource orchestration theory clarifies how firms manage green resources at the network level: structuring involves curating diverse green inputs from network partners to avoid redundancy; bundling transforms these inputs into integrated capabilities; and leveraging applies these capabilities to enhance ESG performance (Sirmon et al., 2011). In parallel, dynamic capabilities explain why some firms succeed in this process: sensing enables firms to detect shifts in green policies or partner resources; seizing facilitates the timely integration of these resources into operations; and reconfiguring ensures alignment with changing institutional demands (Pitelis et al., 2024). Together, these theories address the “resource acquisition–capability transformation–performance realization” chain unique to green innovation networks, where resource value is contingent on both network collaboration and internal adaptive capacity. Recent research by Wang and Zhang (2024) supports this integrated logic. Their study on digital green supply chains shows that inter-organizational cooperation catalyzes eco-innovations through synchronized resource orchestration and deployment of dynamic capabilities. This parallel highlights how collaborative mechanisms in green networks—whether supply chains or

innovation networks—depend on both macro-level resource structuring and micro-level capability adaptation to unlock value. It reinforces the necessity of integrating resource orchestration and dynamic capabilities theories in our framework.

The resource–capability–performance logic embedded in resource orchestration theory provides a solid theoretical foundation for examining the relationship among green innovation network embeddedness, green dynamic capabilities and ESG performance. Within this framework, dynamic capabilities theory offers a process-oriented perspective that systematically explains how resources are acquired, transformed and reconfigured (Cavusgil and Deligonul, 2024). It is particularly effective for elucidating how green resources are converted into ESG performance. Dynamic capabilities primarily include sensing, seizing and reconfiguring. Sensing and seizing reflect a firm's ability to identify external opportunities related to green technologies, policies and institutional changes, and to integrate these opportunities into strategic planning. In traditional contexts, firms typically rely on managerial experience, policy analysis and path dependence to sense opportunities. However, under the green transformation agenda, green resources are increasingly diverse, institutionalized and interactive, making traditional sensing mechanisms insufficient. A firm's ability to identify, filter and convert fragmented and unstructured green resources depends on the presence of systematic green dynamic capabilities.

Green dynamic capabilities refer to a firm's organizational capacity to identify, integrate and reconfigure green technologies, knowledge, information and collaboration mechanisms to build a green competitive advantage (Maksimov et al., 2019). This includes not only the ability to acquire and absorb green technologies, but also the capacity to manage green knowledge flows, adapt to green institutions and coordinate ESG performance outcomes (Mondal et al., 2025). As green innovation networks become a critical platform for accessing green resources, green dynamic capabilities serve as the key link between network embeddedness and ESG performance. Based on resource orchestration theory, this study proposes that green dynamic capabilities mediate the relationship between green innovation network embeddedness and ESG performance.

Moreover, the reconfiguring component of dynamic capabilities emphasizes a firm's ability to continuously renew its management models, organizational structures, and decision-making paradigms in response to external changes (Cavusgil and Deligonul, 2024). The effectiveness of this capability depends on the responsiveness and coordination efficiency of internal systems. In contexts where green network resources are highly fluid and heterogeneous, and rapid decision-making is required, firms that fail to align their strategic intent with organizational structures in a timely manner are less likely to translate green resources into performance outcomes. Organizational agility refers to a firm's capacity to sense environmental complexity, respond swiftly and dynamically adjust its organizational structures, processes and management systems (Rabal-Conesa et al., 2021). Compared with traditional organizational flexibility, agility places greater emphasis on speed and strategic focus, which makes it particularly suitable for environments characterized by high uncertainty and interdependence (Puriwat and Hoonsoopon, 2021). When embedding into green innovation networks, firms with greater organizational agility can better detect the needs of green partners and institutional changes, promptly adjust internal decision-making mechanisms and resource allocation strategies and thereby improve the efficiency of green resource utilization and the development of dynamic capabilities. Therefore, organizational agility may strengthen the effect of green innovation network embeddedness on green dynamic capabilities and amplify the overall impact of network embeddedness on ESG performance.

In summary, this study builds on resource orchestration theory and the dynamic capability perspective to construct a mediation mechanism in which green innovation network embeddedness influences ESG performance through green dynamic capabilities. It also introduces

organizational agility as a moderating variable, thereby developing an integrated analytical framework that incorporates both mediation and moderation effects (see Fig. 1).

2.2. Green innovation network embeddedness and ESG performance

Building on the preceding analysis, this study defines green innovation networks as the formal and informal relational structures formed through ongoing collaboration between enterprises and diverse green actors, including green technology providers, environmental consulting agencies, clean energy platforms, policy advocacy organizations and green finance intermediaries. These networks facilitate the flow, sharing and reconfiguration of green resources across organizational boundaries, serving as a critical external platform for acquiring environmental technologies and sustainability-related knowledge. According to network embeddedness theory (Granovetter, 1985), green innovation network embeddedness has two dimensions: structural embeddedness and relational embeddedness. Structural embeddedness reflects a firm's positional characteristics and degree of connectivity within the network, while relational embeddedness focuses on trust levels and interaction quality between firms and their green partners. This section explores how these two dimensions influence ESG performance in energy enterprises and proposes the relevant hypotheses.

2.2.1. Structural embeddedness and ESG performance

In the context of green transformation, energy enterprises are gradually extending their collaborative boundaries from traditional resource networks to green innovation networks dominated by technologies, institutions and knowledge. As structural embeddedness increases, firms occupy more central positions in the green resource flow, enhancing their bargaining power and access to key environmental resources. Resource orchestration theory suggests that structuring resource portfolios contributes to performance improvement (Carnes et al., 2017). Moderate levels of structural embeddedness enable firms to identify and integrate diverse green resources, build systematic green capabilities and thereby improve ESG performance.

Green innovation networks contain heterogeneous resources such as environmental technologies, low-carbon processes and clean energy solutions. However, the diversity of actors and the decentralized distribution of these resources often result in fragmented information. Firms often struggle to achieve integration benefits through a single channel. Moderate structural embeddedness allows firms to establish effective links with green platforms, universities and standards-setting institutions, transforming fragmented knowledge into coherent solutions aligned with strategic objectives (Ranganathan et al., 2025). For example, in green production, access to carbon emission data, industry-specific ESG metrics and certification information enables firms to identify ESG risks and assess performance with greater precision (Zhou et al., 2024). In addition, moderate structural embeddedness provides access to diverse technical support, such as smart green manufacturing, carbon footprint tracking and digital carbon platforms—which helps achieve environmental goals (Zhang et al., 2024). Since green technologies and standards are subject to regulatory changes, firms that can quickly adapt to updated policies and carbon trading rules may develop dynamic institutional adaptability and enhance governance effectiveness (Teeter and Sandberg, 2017). Furthermore, structural connectivity offers firms opportunities to participate in “virtual green co-creation organizations,” enabling flexible engagement in joint project development, supply chain coordination, and ESG information disclosure, thus enhancing their responsiveness through multilateral collaboration (Yin and Zhao, 2024).

Nevertheless, the value of green resources is highly dependent on their alignment with internal capabilities, mechanisms, and business scenarios. Excessive structural embeddedness may lead to resource overload and green configuration failures, ultimately reducing ESG performance. Notably, these adverse effects are more pronounced in

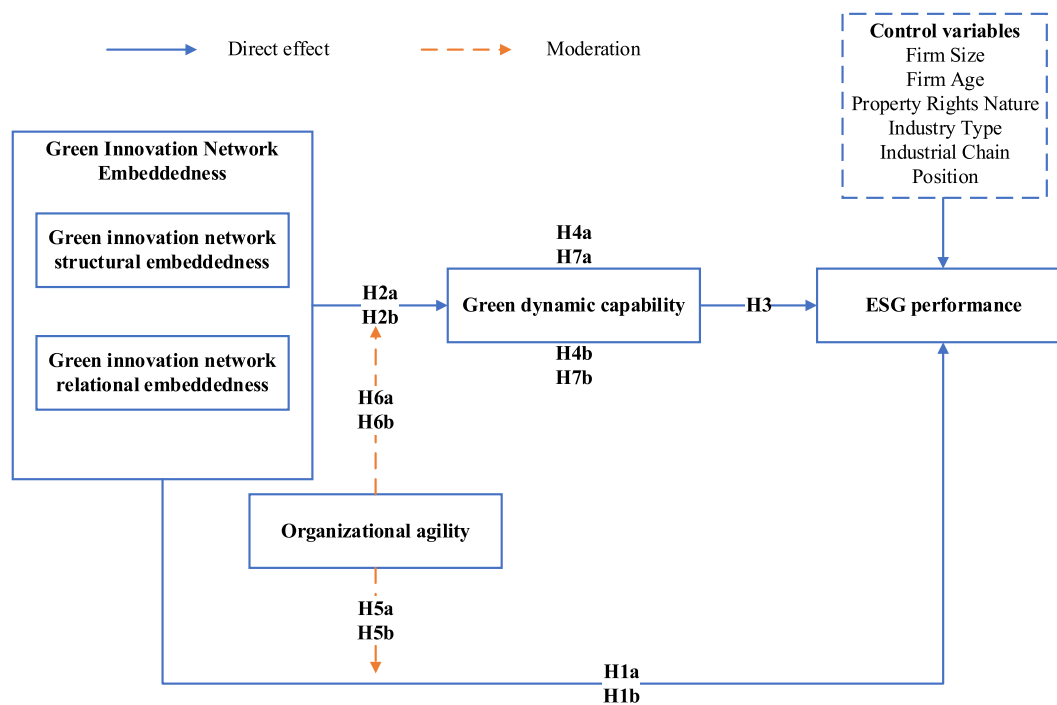


Fig. 1. Proposed theoretical model.

green innovation networks compared to traditional innovation networks due to their unique characteristics. Green innovation networks involve multi-actor, cross-sectoral collaborations—including government regulators, ESG platforms, research institutions, and environmental advocacy groups—each with heterogeneous priorities. This institutional complexity increases coordination friction; when firms become over-embedded, they face heightened risks of value misalignment, decision fatigue, or political entanglement, which magnify the inefficiencies of over-participation. Furthermore, green resources such as carbon emission data, ESG certification tools, and low-carbon models are highly codified but weakly standardized across network actors. Their effective integration requires significant internal adaptation capacity, and excessive inputs may quickly exceed organizational absorption thresholds, leading to fragmented systems and wasted resources. When structural embeddedness becomes too high, firms often adopt multiple green information systems, environmental management technologies, and institutional support platforms. If these systems are incompatible, information silos may emerge, raising integration difficulties and coordination costs (Wang et al., 2024). To achieve system interconnectivity, firms are required to invest in highly specific and sunk-cost-intensive green infrastructures, such as carbon data collection devices, environmental monitoring equipment, and green ERP systems. If these investments are poorly aligned with operational needs, they may crowd out essential resources for substantive ESG improvements (Zhang et al., 2024). Additionally, high embeddedness may reinforce a firm's network position but may also result in internal power imbalances, where green affairs departments gain preferential access to resources, squeezing out investments in other ESG domains such as environmental governance, social responsibility, and corporate governance (Zhu et al., 2025). Thus, we propose the following hypothesis:

H1a. Green innovation network structural embeddedness and ESG performance follow an inverted U-shaped relationship.

2.2.2. Relational embeddedness and ESG performance

Traditional innovation network studies grounded in trust theory argue that higher levels of relational embeddedness enhance trust among partners, facilitate the exchange of specialized resources, private

knowledge, and tacit experience, and reduce transaction costs (McFadyen and Cannella, 2004). Relational embeddedness mitigates information asymmetries and opportunistic behavior, thereby improving resource allocation efficiency (Zhao et al., 2021). However, this logic faces limitations in green innovation networks. Green technologies and knowledge are often disseminated through open standards, cross-organizational platforms, and policy-oriented resources, which are characterized by generalizability and openness (Sahoo et al., 2023). Green resources are highly codified, institutionalized, and frequently transmitted, thereby weakening the exclusivity of relational ties and limiting the explanatory power of the traditional “trust–transaction cost” mechanism (Barbieri et al., 2020).

From the perspective of energy enterprises undergoing green transformation, two key cognitive barriers often emerge. First, firms struggle to shift from a compliance-driven to a strategy-driven understanding of green development. Second, generic green technologies often exhibit weak compatibility with firms' specific operational contexts (Xia et al., 2019). Moderate relational embeddedness helps firms overcome these barriers. Through sustained collaboration with green partners such as technology companies, environmental organizations and industry associations, firms engage in deep interaction at the levels of knowledge, values and practices. Over time, they internalize green governance logic, policy trends and low-carbon pathways, enabling managers to move from reactive compliance to proactive strategy and embedding green values into ESG performance management (Tang et al., 2025). Moreover, relational embeddedness reflects a “value co-creation” orientation rather than a purely transactional logic. Under complex technological and institutional conditions, collaboration involves co-defining governance objectives, designing low-carbon solutions and optimizing production and disclosure processes (Huang et al., 2024). Moderate relational embeddedness enables precise alignment between green technologies and governance scenarios, playing a critical role in the “embedding–transformation–performance” chain.

However, excessive relational embeddedness may suppress ESG performance. High levels of embeddedness may foster redundant trust and reciprocal obligations, reducing information filtering efficiency and crowding out investments in green capability development (Zhao et al., 2025). Persistent embeddedness may reinforce path dependence and

induce technological or institutional lock-in, limiting diversified and adaptive innovation (McFadyen and Cannella, 2004). More importantly, excessive relational embeddedness may lead to overestimating of resource control and inflating green ambitions due to strengthened green discourse power and public pressure. If firms allocate financial and human capital to poorly matched technologies or transformation projects, core ESG capabilities, such as disclosure quality, social responsibility indicators and governance transparency, may be undermined (Sang et al., 2025). Institutional spillover effects may also trigger strategic overextension, shifting investments from capability building to high-risk transformations such as green servitization or carbon financialization, thereby crowding out core ESG efforts (Sui et al., 2024). Thus, we propose the following hypothesis:

H1b. Green innovation network relational embeddedness and ESG performance follow an inverted U-shaped relationship.

2.3. The mediating role of green dynamic capabilities

Green dynamic capabilities refer to a firm's higher-order organizational capabilities to identify, integrate, and reconfigure green resources in the process of pursuing sustainable development (Maksimov et al., 2019). These capabilities enable firms to respond to environmental changes, adapt to institutional pressures and foster green innovation, thereby enhancing green performance. As a specific manifestation of dynamic capabilities, green dynamic capabilities reflect a firm's core competence in sensing opportunities and mobilizing resources during green transitions.

2.3.1. Green innovation network embeddedness and green dynamic capabilities

The development of green dynamic capabilities is jointly influenced by technological access channels, strategic organizational adjustment and the external institutional environment (Mondal et al., 2025). Based on the "resource–capability" framework of resource orchestration theory, this study posits that green innovation network embeddedness fosters the emergence of green dynamic capabilities across technological, organizational and institutional dimensions. However, this influence is not linear, because moderate versus excessive embeddedness may lead to divergent outcomes.

On the technological dimension, structural embeddedness connects firms with a diverse set of green actors, exposing them to a wide range of green technologies, methodologies and solutions. Through connections with green technology platforms, environmental service providers and research institutions, firms gain access to novel and generic green knowledge and technical processes that underpin capability development (Fernando et al., 2019). Relational embeddedness, through in-depth cooperation and experiential sharing, enables firms to acquire tacit knowledge and contextualize technological solutions (Gordano et al., 2024). Frequent interactions and joint practices help firms gradually refine their green capability systems and enhance their technological integration abilities (Juo and Wang, 2022). For example, long-term collaboration with green technology suppliers may facilitate the internalization of green processes and foster systematic capability building. On the organizational dimension, executive support and internal readiness for change are key enablers of green dynamic capabilities (Ullah et al., 2024). Structural embeddedness expands a firm's green collaborative network, broadening managers' strategic perspectives and increasing their willingness to invest in green transitions (Caldera et al., 2019). Simultaneously, relational embeddedness strengthens alignment with core green partners, fostering coherence between environmental goals and capability-building efforts. For instance, through close engagement with leading green firms, companies may adopt best practices in green management, optimize internal resource allocation and reinforce the organizational foundation for capability development. On the institutional dimension, higher levels of

structural embeddedness expose firms to more stringent regulatory pressure and greater public scrutiny. The expansion of green partner networks amplifies institutional expectations, motivating firms to build green capabilities to mitigate compliance risks and safeguard reputational assets (Baah et al., 2020; Shah, 2011). Relational embeddedness opens channels for accessing institutional resources such as project funding, certification support and policy interpretation (Sadovnikova and Pujari, 2017). For example, close ties with industry associations allow firms to obtain timely green policy updates, facilitating capability planning and development.

Nevertheless, more embeddedness is not always better. This non-linearity is particularly striking in green innovation networks. Excessive structural embeddedness may lead to the firm connecting with too many sources of green technologies that lack compatibility, forming technological silos that hinder integration (Peng et al., 2020b). Unlike traditional technology networks with standardized knowledge bases, green technologies (e.g., carbon capture systems, renewable energy solutions) vary widely in technical principles and application scenarios, making cross-source integration for more complex. Excessive relational embeddedness may result in knowledge homogeneity and cognitive lock-in, limiting the diversity and evolution of capability systems (Yongsheng et al., 2021). In green contexts, this lock-in is risky because environmental policies and stakeholder expectations evolve rapidly; firms overly reliant on a narrow set of relational partners may fail to adapt to new green standards. Organizationally, high embeddedness may cause managerial overreliance on external network positions, leading to underinvestment in internal capabilities (Moran, 2005). Institutionally, excessive structural embeddedness increases coordination costs in resource integration and may constrain the firm when adjusting its capability paths (Zhou et al., 2020). Excessive relational embeddedness may intensify perceived external pressure, prompting irrational technology adoption or strategic distraction, ultimately weakening capability development. Thus, we propose the following hypotheses:

H2a. Green innovation network structural embeddedness and green dynamic capabilities follow an inverted U-shaped relationship.

H2b. Green innovation network relational embeddedness and green dynamic capabilities follow an inverted U-shaped relationship.

2.3.2. Green dynamic capabilities and ESG performance

According to resource orchestration theory, dynamic capabilities enable value creation by guiding resource integration and opportunity recognition (Sirmon et al., 2011). As a core capability in corporate sustainability transitions, green dynamic capabilities allow firms to respond to regulatory changes, technological evolution, and stakeholder expectations. Their role lies in the dynamic allocation of green resources to ensure stable and forward-looking ESG performance.

This study posits that firms with stronger green dynamic capabilities are better positioned to identify and integrate environmental, social, and governance-related information and, through systematic analysis and capability accumulation, execute ESG strategies more precisely (Maksimov et al., 2019). On the environmental dimension, such firms can promptly access carbon emission data, energy efficiency metrics, and green certification information, which they embed into product design, production processes, and clean technology application, continuously improving environmental outcomes (Gao et al., 2024; Uddin and Rahman, 2012; Woo et al., 2021). For instance, with forward-looking technology reserves, firms can anticipate shifts in green standards and proactively mitigate environmental risks. On the social dimension, green dynamic capabilities enhance a firm's sensitivity to stakeholders' expectations, such as employee well-being and community engagement (Yuan and Cao, 2022). By coordinating resources, firms embed social responsibility into corporate culture and operational processes, thereby enhancing social performance (Zhang et al., 2022). For example, firms using ESG analytics tools may identify labor rights issues

early and optimize governance paths to build public trust. On the governance dimension, green dynamic capabilities support improvements in information disclosure and organizational structure. Firms with strong capabilities can quickly align with regulatory demands and market preferences, producing high-quality ESG reports that boost transparency. They may also promote institutional adaptation, such as establishing ESG task forces or enhancing board oversight. These efforts improve governance performance and may help firms gain advantages in green finance and ESG ratings (Yuan and Cao, 2022). Thus, we propose the following hypothesis:

H3. Green dynamic capabilities are positively associated with ESG performance.

2.3.3. The mediating role and nonlinear nature of green dynamic capabilities

Following the logic of “resource acquisition–capability formation–performance improvement” in resource orchestration theory (Xie et al., 2024b), green dynamic capabilities play a critical mediating role between green innovation network embeddedness and ESG performance. Moderate embeddedness offers diverse green resources that accelerate capability development, thereby enhancing ESG performance. When structural embeddedness is moderate, firms experience regulatory pressure and peer demonstration effects that heighten their motivation to acquire resources and build capabilities (Aragón-Correa et al., 2020; Liao, 2018). When relational embeddedness is moderate, trust mechanisms encourage green partners to provide technical guidance and institutional support, facilitating capability development.

During this process, managerial cognition of green transition deepens, and firms shift from passive compliance to capability-driven strategies (Wu et al., 2025). This transition enables firms to enhance ESG performance through environmental, social, and governance-related capabilities. However, excessive embeddedness may weaken this mediating effect. When structural embeddedness is too high, resource redundancy and fragmented technologies may form isolated green information silos, hindering capability development and limiting ESG gains. When relational embeddedness is too high, dependence on external green resources intensifies. Under multiple institutional pressures, managers may overreact, investing excessively in green capabilities with diminishing returns. Such deviations in capability formation may divert firms from their strategic focus in pursuing ESG goals and reduce performance returns. For example, failure to integrate green technologies or misallocation of governance resources in an overly embedded network may impair long-term competitiveness. Thus, we propose the following hypotheses:

H4a. Green dynamic capabilities mediate the relationship between green innovation network structural embeddedness and ESG performance in a nonlinear manner.

H4b. Green dynamic capabilities mediate the relationship between green innovation network relational embeddedness and ESG performance in a nonlinear manner.

2.4. The moderating role of organizational agility

Organizational agility refers to a firm’s ability to rapidly adjust its organizational structure, process configuration, and resource allocation in response to changes in internal and external environments or strategic goals, thereby achieving efficient responsiveness and dynamic adaptability (Rabal-Conesa et al., 2021). As a key manifestation of reconfiguration capability, organizational agility reflects whether a firm can detect signals of change in complex environments and engage in real-time organizational redesign to unlock resource potential and enhance strategic execution.

2.4.1. The moderating role of organizational agility between green innovation network embeddedness and ESG performance

Under conditions of moderate structural embeddedness, firms establish connections with green actors to access green technologies, environmental policies, and governance tools. When organizational agility is high, firms can quickly restructure internal systems, create green project teams, and adjust business processes to deeply integrate green resources with existing operations at low cost, thus enhancing ESG performance (Mariani et al., 2025). In addition, highly agile firms typically possess strong inter-organizational collaboration capabilities (Guo et al., 2023), enabling them to establish green co-creation platforms through high-frequency coordination mechanisms, which further amplify the positive effects of structural embeddedness on ESG performance. However, under conditions of excessive structural embeddedness, high organizational agility may exacerbate the negative effects of over-embeddedness. On one hand, agility grants firms rapid resource mobilization capabilities, which can lead to accelerated investment in green infrastructure that crowds out investment in other ESG domains, ultimately undermining overall performance (Wu et al., 2025). On the other hand, agility may concentrate attention and resource allocation on green affairs departments, forming internal “power centers” that suppress the development of other governance functions, thereby disrupting ESG integration and lowering comprehensive performance.

Notably, the moderating effect of organizational agility is weaker in the structural embeddedness pathway due to the inherently topological and resource-access nature of structural embeddedness. Structural embeddedness primarily reflects positional advantages and resource diversity, which are relatively stable and path-dependent. These attributes are less sensitive to short-term organizational adjustments or dynamic reconfiguration capabilities—key features of agility. In contrast, relational embeddedness is rooted in trust, reciprocal interaction, and shared norms, creating contexts where agile sensing and adaptation are more impactful. In moderately embedded relational networks, high organizational agility helps firms build flexible talent orchestration and collaboration mechanisms, effectively managing the challenges of cross-organizational resource deployment and project matching (Mao et al., 2024). Agile firms can rapidly align their internal structures and governance systems with the resources, methods, and governance logics provided by green partners, thus enhancing the efficiency of green governance and ESG performance. However, under excessive relational embeddedness, firms may place irrational trust in external resources, misjudge the scope and value of green initiatives, and make strategic missteps. High agility reinforces these risks by enabling swift execution and resource allocation, which can lead to short-term imbalances in ESG investment and greater volatility or even decline in green performance. Thus, we propose the following hypotheses:

H5a. Organizational agility positively moderates the inverted U-shaped relationship between green innovation network structural embeddedness and ESG performance, such that the curve becomes steeper when organizational agility is high.

H5b. Organizational agility positively moderates the inverted U-shaped relationship between green innovation network relational embeddedness and ESG performance, such that the curve becomes steeper when organizational agility is high.

2.4.2. The moderating role of organizational agility between green innovation network embeddedness and green dynamic capabilities

Under moderate structural embeddedness, firms benefit from regulatory pressure and resource supply arising from network participation, which drives the development of green dynamic capabilities (Singh et al., 2021). When organizational agility is high, firms can flexibly adjust structures, establish green management centers or ESG departments, and rapidly align internal systems with external green resource flows, thereby facilitating structured capability building. Agility also enhances the flexibility of resource allocation, enabling

firms to integrate projects, personnel, and financial inputs more efficiently during capability construction, thereby reinforcing the effect of embeddedness on capability development. However, under conditions of excessive structural embeddedness, firms may be inclined to outsource parts of green integration tasks or rely on dominant firms' green standards and technological pathways, which weakens their internal motivation for capability development. In this context, agility may accelerate the execution of path-dependent strategies, intensifying capability lock-in and reducing diversity in capability evolution (Yongsheng et al., 2021), thereby amplifying the negative effects of structural embeddedness.

The limited moderating effect of organizational agility on structural embeddedness and green dynamic capabilities stems from the fact that structural embeddedness focuses on resource access rather than interactive learning. Green dynamic capabilities, particularly those related to technological integration and institutional adaptation, depend more on stable resource flows than on agile reconfiguration in structurally embedded contexts. In contrast, relational embeddedness thrives on iterative interaction and knowledge co-creation, where agility can enhance the speed and accuracy of capability development. In the case of moderate relational embeddedness, high organizational agility allows firms to rapidly identify and utilize institutional, experiential, and platform-based support resources provided by partners, while dynamically aligning organizational structures with green capability strategies, thus accelerating the pace of capability formation. In contrast, low-agility firms often display organizational inertia, making them slow to respond to institutional changes or tool-based transformation demands raised by partners, thereby weakening the positive effects of relational embeddedness (Omidvar et al., 2022). However, under excessive relational embeddedness, green cognition between firms and partners tends to converge, and knowledge becomes homogenized (Zhao et al., 2025). Agile firms, due to their rapid responsiveness, may overcommit resources to a single green path, which intensifies strategic imbalance, narrows the range of green resource recognition and integration, and reinforces the adverse effects of over-embeddedness. Thus, we propose the following hypotheses:

H6a. Organizational agility positively moderates the inverted U-shaped relationship between green innovation network structural embeddedness and green dynamic capabilities, such that the curve becomes steeper when organizational agility is high.

H6b. Organizational agility positively moderates the inverted U-shaped relationship between green innovation network relational embeddedness and green dynamic capabilities, such that the curve becomes steeper when organizational agility is high.

2.4.3. The moderated mediation role of organizational agility

In the process of developing green dynamic capabilities, firms often need to flexibly adjust organizational structures and process mechanisms to enhance the efficiency of resource integration and the speed of capability formation. Actions such as establishing green governance centers, ESG implementation departments, or cross-functional green collaboration mechanisms represent typical organizational responses to embedded green collaboration. Firms with high agility can rapidly adjust internal structures and allocate resources, creating capability-receiving mechanisms that strengthen the link between green innovation network embeddedness and green dynamic capability formation. As proposed in Hypotheses H4a and H4b, moderate levels of green innovation network embeddedness enhance ESG performance by fostering green dynamic capabilities. However, excessive embeddedness may hinder capability formation due to resource redundancy, path dependence, and coordination failure, thereby weakening sustainable strategic outcomes (Wang and Zhang, 2025a). Building on this logic, this study further posits that the moderating effect of organizational agility on the relationship between green innovation network embeddedness and green dynamic capabilities indirectly influences ESG performance,

resulting in a moderated mediation effect. Specifically, when organizational agility is high, firms can promptly and accurately respond to internal alignment needs during green resource inflow, accelerating the internalization and transformation of green capabilities and amplifying the "blessing" effects of green embeddedness on ESG performance. However, agility may also intensify structural distortions during the embedding process, such as biased investment in green initiatives or entrenched path dependence, thereby exacerbating the "curse" effects. Therefore, organizational agility not only shapes the efficiency of converting green embeddedness into capabilities but also indirectly affects the final ESG performance outcomes through capability formation. Thus, we propose the following hypotheses:

H7a. The moderating effect of organizational agility on the inverted U-shaped relationship between green innovation network structural embeddedness and ESG performance is mediated by green dynamic capabilities.

H7b. The moderating effect of organizational agility on the inverted U-shaped relationship between green innovation network relational embeddedness and ESG performance is mediated by green dynamic capabilities.

3. Research methodology

3.1. Variable measurement

To ensure both scientific rigor and empirical reliability, this study uses well-established and validated measurement scales from authoritative academic literature for all core variables. For constructs that are still evolving and lack mature measurement systems, such as green innovation network embeddedness, we adapted and refined scale items based on existing theoretical insights and field-based observations to improve contextual relevance and measurement accuracy.

Green Innovation Network Embeddedness (GINE). Following Granovetter's (1985) network embeddedness framework and drawing on the contextual insights from Yuan and Cao (2022), this study conceptualizes GINE through two dimensions: structural embeddedness and relational embeddedness. To ensure the scale's validity and relevance for energy enterprises in Northwestern China, we conducted a two-stage validation process before formal data collection: First, we consulted a panel of six experts was consulted for content validation. The panel comprised three senior ESG managers from regional energy firms (with over 10 years of experience in green transition), two researchers specializing in green supply chain management and one policy adviser from a provincial energy transition task force. Experts assessed each draft item for clarity, contextual fit and relevance to the unique features of green innovation networks in the energy sector—particularly in Northwestern China, where cross-regional coordination, government-led low-carbon demonstration projects and ESG digitalization are key characteristics. Based on their feedback, we revised item wordings to enhance sectoral alignment: for example, adding references to "green platforms for energy transition," "carbon accounting systems" and "energy regulatory institutions" to better reflect local collaboration dynamics. Specific revisions included modifying "green technology organizations" to "green technology providers and carbon management platforms" (GINSE1) and adjusting "cooperation with green partners" to "collaboration with energy-specific green institutions" (GINRE2).

Second, we conducted a pilot test was conducted with 32 energy enterprises in Shaanxi and Gansu provinces (key energy hubs in Northwestern China) to refine the scale. We performed item-total correlation analysis and exploratory factor analysis (EFA) on the pilot data. Items with factor loadings below 0.6 or ambiguous semantic overlap were modified or removed. For example, an original item on "network centrality" was revised to "centrality in regional green energy collaboration networks" to strengthen contextual specificity. The pilot results confirmed strong internal consistency (Cronbach's $\alpha = 0.88$ for

structural embeddedness and 0.89 for relational embeddedness), supporting the retention of the final scale structure. Structural embeddedness captures a firm's breadth of connections, centrality and access to resources within the green innovation network. It is measured using four items that quantify the firm's structural positioning and resource accessibility. Relational embeddedness focuses on the intensity and trust level of collaborations between the firm and green technology providers, environmental research institutions and policy platforms. Six items were designed to assess the quality and depth of these relationships.

Green Dynamic Capabilities (GDC). We assessed GDC using four items adapted from Singh et al. (2021), with contextual revisions to emphasize green-specific competencies relevant to energy enterprises.

ESG Performance (ESGP). ESGP was measured using a 15-item scale comprising three sub-dimensions: environmental performance (five items), social performance (five items), and governance performance (five items), adapted from Gidage and Bhide (2025).

Organizational Agility (OA). Organizational agility was measured using six items adapted from Mariani et al. (2025), capturing a firm's ability to respond rapidly and reconfigure internal processes under changing conditions.

All items were rated on a seven-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree), ensuring data granularity and comparability. To control for potential confounding effects, this study follows Tang et al. (2025) by incorporating firm size, firm age, property rights nature, state-owned enterprise, private enterprise, industry type and industry chain position as control variables.

3.2. Data collection

To ensure the practical relevance and policy value of the research findings, this study focuses on energy enterprises located in North-western China. This region represents a strategic frontier of the "Western Development" and "Dual Carbon" initiatives and serves as a critical area for the transformation of traditional energy firms into green energy players. The formal data collection was conducted from April to June 2024.

To ensure sufficient sample coverage and data depth, the research team collaborated with numerous energy enterprises, local energy authorities, industry associations and green technology service providers to build a research network covering more than 1200 firms. This partnership ensured smooth survey implementation and enhanced the representativeness of the sample. Our sampling strategy employed stratified purposive sampling across four provinces in Northwestern China: Shaanxi, Gansu, Ningxia and Inner Mongolia. The target population was defined as energy enterprises registered in three types of official databases: (1) provincial industrial and commercial registries (covering all energy-related firms with annual revenue >20 million RMB); (2) regional carbon trading participant lists (including firms subject to mandatory emission reporting); and (3) member directories of local energy industry associations. From these databases, we randomly selected 1200 firms, ensuring coverage across sub-sectors and firm sizes. During data collection, we adopted a combined ex-ante and ex-post quality-control strategy. In the ex-ante phase, to reduce information bias and enhance response reliability, respondents were required to be mid- to senior-level managers with at least two years of tenure and in-depth knowledge of the firm's green transition and ESG practices. In the ex-post phase, we applied both manual inspection and automated filters to ensure data validity. Questionnaires with patterned responses, missing values or responses from individuals not meeting the eligibility criteria were excluded.

The survey was conducted electronically and distributed directly to target managers via directed mailing. A total of 602 responses were collected. After data screening and cleaning, 566 valid responses were retained. As shown in Table 1, the final sample reflects high diversity and structural heterogeneity across dimensions such as firm size, firm age, property rights nature, property rights nature, industry type and

Table 1
Sample characteristics (N = 566).

Variable	Category	Count	Percentage
Firm Size	Large enterprises	219	38.70 %
	Small and medium-sized enterprises	347	61.30 %
Firm Age	5 years or below	102	18.03 %
	Above 5 years and up to 10 years	190	33.57 %
	Above 10 years and up to 15 years	172	30.39 %
	More than 15 years	102	18.03 %
Property Rights Nature	State-owned enterprises	161	28.45 %
	Privately-owned enterprises	318	56.19 %
	Other types of enterprises	87	15.37 %
Industry Type	Strategic emerging industries	300	53.00 %
	Traditional industries	266	47.00 %
Industrial Chain Position	Midstream and downstream enterprises	382	67.63 %
	Upstream enterprises	184	32.37 %

industry chain position. No significant skewness was observed, indicating strong representativeness of the sample. However, we acknowledge that our collaboration with energy authorities, industry associations, and green technology service providers—while improving response rates—may have introduced selection bias: firms already engaged in green initiatives or with higher ESG awareness may be overrepresented, particularly in strategic emerging sectors such as renewable power, energy storage, and environmental equipment manufacturing. Traditional energy sub-sectors are proportionally underrepresented compared to their actual share in China's energy industry.

4. Results

4.1. Reliability and validity tests

Before conducting hypothesis testing, this study systematically evaluated the reliability and validity of the measurement instruments to ensure the robustness of the data and the scientific rigor of the theoretical interpretation. For reliability, Cronbach's α coefficients and composite reliability (CR) values were calculated (Hair et al., 2019). The α values for all constructs ranged from 0.874 to 0.892, and the CR values all exceeded 0.866, well above the commonly accepted threshold of 0.70. Notably, the green innovation network embeddedness scale (both structural and relational dimensions) showed strong internal consistency (Cronbach's $\alpha = 0.881$ for structural embeddedness and 0.874 for relational embeddedness), confirming the effectiveness of the expert validation and pilot test refinements. These results demonstrate high internal consistency and overall reliability, providing a solid empirical foundation for subsequent analyses. Additionally, the standardized factor loadings for all items were statistically significant, indicating stable performance of the measurement items in capturing their corresponding latent constructs.

For validity, this study assessed content validity, convergent validity, and discriminant validity (Gidage et al., 2025). Content validity for green innovation network embeddedness was enhanced through the two-stage expert validation and pilot test, as detailed in Section 3.1. Expert feedback ensured alignment with the unique context of North-western China's energy sector, while pilot test results confirmed item clarity and structural stability. Content validity was ensured by adopting scale items from well-established literature in leading journals and modifying the semantics based on the research context of green innovation and sustainable governance. For convergent validity, confirmatory factor analysis (CFA) was conducted to assess model fit. The results showed that $\chi^2/df = 1.869$, CFI = 0.954, TLI = 0.946, IFI = 0.951, and RMSEA = 0.038. All indices met or exceeded the recommended thresholds, indicating strong model fit. The average variance extracted (AVE) values for all constructs ranged from 0.523 to 0.684, exceeding

the 0.50 benchmark, which confirms convergent validity. The results of the reliability and validity test are shown in [Table 2](#).

To test discriminant validity, several competing models were constructed and compared. The hypothesized model demonstrated superior fit compared to alternative merged or two-factor models ([Table 3](#)). Furthermore, the square root of each construct's AVE exceeded its correlations with other constructs, indicating good construct independence and discriminant validity. In sum, the scales used in this study exhibited strong reliability and validity, providing a sound basis for hypothesis testing.

4.2. Common method variance

Given that the data were collected primarily through self-reported responses from enterprise managers, the potential issue of common method variance (CMV) was addressed using multiple diagnostic procedures ([Simmering et al., 2015](#)). First, Harman's single-factor test was conducted. The unrotated principal component analysis extracted five factors with eigenvalues greater than one, and the first factor accounted for 31.84 % of the total variance, which is below the 40 % threshold, indicating no serious threat from CMV.

To further ensure robustness, the unmeasured latent method factor approach was employed ([Ding et al., 2023](#)). A baseline CFA model (Model M1) was constructed and compared with a method-factor model (Model M2) that included an additional latent method factor. The changes in model fit were $\Delta CFI = 0.011$, $\Delta TLI = 0.010$, $\Delta NFI = 0.015$, and $\Delta RMSEA = 0.005$, all below the critical threshold of 0.02. These results suggest that adding the method factor did not significantly improve model fit, confirming that CMV was effectively controlled and did not substantially bias the findings.

4.3. Descriptive and correlation analysis

Descriptive statistics for all variables indicated reasonable means and standard deviations, with no extreme outliers, confirming the high quality of the data ([Table 4](#)). Correlation analysis revealed that most core variables were significantly correlated at the 0.01 level, except for the relationship between green dynamic capabilities and organizational agility, which was not statistically significant ([Table 5](#)). This finding suggests the potential existence of mediating or moderating variables, which warrants further investigation.

4.4. Test of direct effects

To rigorously test the hypothesized inverted U-shaped relationships, this study employed the three-step testing approach proposed by [Haans et al. \(2016\)](#) to examine the nonlinear effects of green innovation network embeddedness on green dynamic capabilities and ESG performance. First, both structural and relational embeddedness variables were standardized to a mean of zero and a standard deviation of one, and their squared terms were constructed to capture nonlinear effects. Hierarchical regression analysis was then conducted to assess the significance of the linear and non-linear terms. Second, we formulated nonlinear functional equations from the standardized regression coefficients, nonlinear functional equations were formulated and differentiated to obtain expressions for the first derivatives. We examined the slopes at the upper and lower boundaries of each variable's range to assess whether the slope signs followed the "increase-then-decrease" pattern indicative of an inverted U-shape. Third, we calculated the turning point (i.e., the point where the first derivative equals zero) was calculated and compared against the actual range of the independent variable. An inverted U-shaped relationship is confirmed when the regression coefficient of the squared term is significantly negative, the slope at the minimum value is positive, the slope at the maximum value is negative and the turning point lies within the variable's observed range.

Table 2

Measurement items, indicator loadings and the results of CR, Cronbach's α , AVE.

Item	Loading	Source
Green Innovation Network Structural Embeddedness (GINSE)		Cronbach's $\alpha = 0.881$ AVE = 0.711 CR = 0.866
Our firm has established stable connections with various green technology providers, carbon management platforms, and energy policy institutions. (GINSE1)	0.876	Yang et al. (2011); Suki et al. (2022)
Our firm occupies a central and influential position within regional green energy collaboration networks. (GINSE2)	0.854	
We have easy access to green technologies and policy-related resources through the network. (GINSE3)	0.789	
We actively participate in multiple green innovation platforms for energy transition (e. g., wind-solar-hydrogen integration alliances). (GINSE4)	0.821	
Green Innovation Network Relational Embeddedness (GINRE)		Cronbach's $\alpha = 0.874$ AVE = 0.741 CR = 0.889
We maintain long-term and stable cooperative relationships with green technology suppliers specializing in energy sector solutions. (GINRE1)	0.900	Yang et al. (2011); Suki et al. (2022)
We collaborate deeply with energy-specific green institutions on green projects. (GINRE2)	0.841	
Our cooperation with green platforms is built on high levels of trust. (GINRE3)	0.826	
We frequently engage in informal exchanges of experience and knowledge with green partners. (GINRE4)	0.833	
We share a common vision and goals with our green innovation partners regarding low-carbon energy transformation. (GINRE5)	0.856	
Our cooperation mechanisms promote the co-development of green technologies and knowledge in energy fields. (GINRE6)	0.832	
Green Dynamic Capabilities (GDC)		Cronbach's $\alpha = 0.876$ AVE = 0.670 CR = 0.922
We are able to identify external technologies and resources valuable for green transformation. (GDC1)	0.973	Mensah et al. (2025); Yousaf (2021)
We effectively integrate acquired green resources into our operations. (GDC2)	0.794	
We can adjust our green development paths in response to external changes. (GDC3)	0.785	
We reconfigure our green capabilities according to policy and technological evolution. (GDC4)	0.782	
ESG Performance (ESGP)		Cronbach's $\alpha = 0.878$ AVE = 0.574 CR = 0.874
We continuously reduce energy and resource consumption. (ESGP1)	0.984	Gidage and Bhide (2025)
We implement initiatives to reduce carbon emissions. (ESGP2)	0.703	
We adopt green processes and clean production technologies. (ESGP3)	0.712	
We achieve progress in green product development. (ESGP4)	0.709	
We actively disclose environmental information. (ESGP5)	0.711	
We provide a good working environment and training opportunities for employees. (ESGP6)	0.723	
We value feedback and engagement from customers and communities. (ESGP7)	0.834	
We prioritize product safety and consumer rights. (ESGP8)	0.843	
We support public welfare and social responsibility programs. (ESGP9)	0.822	
We ensure employee health and well-being. (ESGP10)	0.862	
Our governance structure supports sustainable development goals. (ESGP11)	0.851	
We have transparent mechanisms for environmental and social information disclosure. (ESGP12)	0.847	
We comply with all relevant environmental and sustainability regulations. (ESGP13)	0.863	

(continued on next page)

Table 2 (continued)

Item	Loading	Source
We have dedicated senior leadership or committees responsible for green development. (ESGP14)	0.837	
Our board of directors emphasizes corporate sustainability strategies. (ESGP15)	0.711	
Organizational Agility (OA) Cronbach's $\alpha = 0.892$	AVE = 0.586	CR = 0.911
We can quickly respond to changes in green policies and markets. (OA1)	0.821	Mikalef and Pateli (2017)
We can flexibly adjust our organizational structure to support green transition. (OA2)	0.842	
We have agile cross-departmental coordination mechanisms. (OA3)	0.834	
We can rapidly allocate resources to support green initiatives. (OA4)	0.822	
Our management processes are flexible and responsive. (OA5)	0.846	
Our strategic adjustments are faster than the industry average. (OA6)	0.844	

Table 3

Confirmatory factor analysis: Discriminant validity test (N = 566).

Model	Factors	Chi-square/ df	CFI	TLI	IFI	RMSEA
Model A	5 factors: GINSE; GINRE; GDC; ESGP; OA	1.869	0.954	0.946	0.951	0.038
Model B	4 factors: GINSE + GINRE; GDC; ESGP; OA	2.403	0.921	0.916	0.921	0.05
Model C	1 factor: GINSE + GINRE + GDC + ESGP + OA	8.299	0.586	0.561	0.587	0.114

Note: GINSE = Green Innovation Network Structural Embeddedness; GINRE = Green Innovation Network Relational Embeddedness; GDC = Green Dynamic Capabilities; ESGP = ESG Performance; OA = Organizational Agility.

As shown in Table 6, in Models 2 and 6, the squared term of structural embeddedness had significant negative effects on green dynamic capabilities ($\beta = -0.121$, $p < 0.001$) and ESG performance ($\beta = -0.083$, $p < 0.001$), indicating the presence of significant nonlinear effects. In Models 3 and 8, the squared term of relational embeddedness also showed significant negative impacts on green dynamic capabilities ($\beta = -0.129$, $p < 0.001$) and ESG performance ($\beta = -0.104$, $p < 0.001$), meeting the first condition for an inverted U-shape. To evaluate the second condition, we calculated the first derivatives of the functional forms were calculated (Table 7). The standardized value ranges for structural and relational embeddedness were $[-2.652, 2.015]$ and $[-2.931, 1.964]$, respectively. For both dimensions, the slope at the minimum value was positive and at the maximum value negative,

Table 4

Descriptive statistical table of main variables.

Variable	N	Mean	Std. Dev.	Min	P25	Median	P75	Max
GINSE	566	4.472	1.211	1.000	3.800	4.500	5.200	6.900
GINRE	566	4.598	1.144	1.200	4.000	4.650	5.450	7.000
OA	566	4.534	0.934	1.500	4.500	4.667	5.167	6.400
GDC	566	4.279	1.082	1.000	3.625	4.410	5.000	6.710
ESGP	566	4.382	1.096	1.300	3.600	4.480	5.278	6.570
Size	566	0.387	0.488	0.000	0.000	0.000	1.000	1.000
Age	566	2.482	0.961	1.000	2.000	2.000	3.000	4.000
SOE	566	0.285	0.452	0.000	0.000	0.000	1.000	1.000
PE	566	0.563	0.497	0.000	0.000	1.000	1.000	1.000
IT	566	0.530	0.500	0.000	0.000	1.000	1.000	1.000
ICP	566	0.675	0.468	0.000	0.000	1.000	1.000	1.000

Note: GINSE = Green Innovation Network Structural Embeddedness; GINRE = Green Innovation Network Relational Embeddedness; GDC = Green Dynamic Capabilities; ESGP = ESG Performance; OA = Organizational Agility.

satisfying the second condition. Analysis of turning points confirmed that all four fell within their respective standardized ranges, satisfying the third condition. Thus, Hypotheses H1a, H1b, H2a, and H2b were supported, confirming that both structural and relational embeddedness exhibit inverted U-shaped relationships with green dynamic capabilities and ESG performance. Additionally, Model 5 showed that green dynamic capabilities had a significant positive effect on ESG performance ($\beta = 0.397$, $p < 0.001$), supporting Hypothesis H3. These findings suggest that moderate levels of green innovation network embeddedness enhance both capability development and ESG performance, while excessive embeddedness may result in resource redundancy or coordination failure, thereby diminishing expected benefits.

4.5. Test of nonlinear mediation effects

To further explore the mediating role of green dynamic capabilities, we applied a three-step procedure to assess nonlinear mediation effects. As shown in Table 5, both the linear and squared terms of structural and relational embeddedness significantly influenced green dynamic capabilities and ESG performance. In the full models, green dynamic capabilities continued to exert a significant positive effect on ESG performance. Specifically, in Model 7, after including structural embeddedness (linear and squared terms) and green dynamic capabilities, the coefficient for green dynamic capabilities remained significant ($\beta = 0.325$, $p < 0.001$). Similarly, in Model 9, after including relational embeddedness and green dynamic capabilities, the coefficient remained significant ($\beta = 0.251$, $p < 0.001$).

We used the Medcurve macro developed by Hayes and Preacher (2010) and applied 5000 bootstrap resamples to test the nonlinear mediation effects. For structural embeddedness, the indirect effects of green dynamic capabilities on ESG performance at one standard deviation below the mean, at the mean and one standard deviation above the mean were 0.187, 0.101, and 0.024, respectively. The 95 % confidence intervals were (0.125, 0.263), (0.068, 0.147) and $(-0.028, 0.079)$, respectively. The first two intervals did not include zero, indicating significant mediation at low to moderate levels of embeddedness, but no significant mediation at high levels. For relational embeddedness, the indirect effects at the three respective levels were 0.153, 0.085, and 0.019, with 95 % confidence intervals of (0.094, 0.222), (0.052, 0.127), and $(-0.020, 0.065)$. Again, significant mediation was observed at low to moderate levels but none at high levels.

These indirect effects represent standardized effect sizes, reflecting the practical magnitude of the mediating role of green dynamic capabilities across different embeddedness levels. For structural embeddedness, the 0.187 indirect effect at low embeddedness implies that a one-standard-deviation increase in structural embeddedness is associated with a 0.187-standard-deviation improvement in ESG performance via green dynamic capabilities—equivalent to approximately a 12 % increase in raw ESG scores (based on the sample's ESG performance standard deviation of 1.096). The decline to 0.101 at moderate

Table 5
Correlation matrix of main variables.

Variable	Size	Age	SOE	PE	IT	ICP	GINSE	GINRE	GDC	OA	ESGP
Size											
Age	0.654**										
SOE	−0.008	0.007									
PE	0.020	−0.002	−0.726**								
IT	0.162**	−0.079*	−0.069	0.035							
ICP	−0.017	0.011	0.031	0.047	−0.058						
GINSE	0.077	0.046	−0.068	0.082	0.088*	0.051	0.843				
GINRE	0.091*	0.029	−0.130**	0.141**	0.129**	0.017	0.640**	0.861			
GDC	0.133**	0.085*	−0.107*	0.072	0.210**	−0.038	0.408**	0.460**	0.819		
OA	0.073*	0.099*	−0.112*	0.110*	0.090	0.089*	0.122**	0.166**	0.052	0.766	
ESGP	0.174**	0.139**	−0.197**	0.160**	0.122**	0.060	0.326**	0.310**	0.441**	0.443**	0.758

Note: ** $p < 0.01$, $p < 0.05$ (two-tailed). Diagonal bolded values represent the square root of the AVE for each construct, indicating discriminant validity. GINSE = Green Innovation Network Structural Embeddedness; GINRE = Green Innovation Network Relational Embeddedness; GDC = Green Dynamic Capabilities; ESGP = ESG Performance; OA = Organizational Agility.

Table 6
Hierarchical regression analysis results (N = 566).

Variable	GDC				ESGP				
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Firm Size	0.198 (0.110)	0.129 (0.102)	0.071 (0.100)	0.251* (0.109)	0.162 (0.101)	0.198 (0.106)	0.149 (0.099)	0.122 (0.101)	0.137 (0.097)
Firm Age	0.011 (0.055)	0.030 (0.051)	0.061 (0.054)	0.057 (0.061)	0.049 (0.056)	0.065 (0.057)	0.056 (0.059)	0.097* (0.050)	0.079 (0.048)
SOE	−0.248 (0.131)	−0.299* (0.132)	−0.195 (0.125)	−0.321* (0.122)	−0.222 (0.119)	−0.308** (0.123)	−0.234* (0.128)	−0.267* (0.117)	−0.240* (0.124)
PE	−0.066 (0.120)	−0.101 (0.116)	−0.112 (0.112)	0.105 (0.118)	0.103 (0.111)	0.084 (0.115)	0.083 (0.109)	0.081 (0.107)	0.079 (0.104)
IT	0.345*** (0.083)	0.298*** (0.076)	0.255*** (0.074)	0.179* (0.082)	0.038 (0.076)	0.141 (0.078)	0.042 (0.075)	0.093 (0.073)	0.026 (0.072)
ICP	−0.057 (0.088)	−0.076 (0.080)	−0.045 (0.078)	0.125 (0.087)	0.133 (0.083)	0.087 (0.085)	0.126 (0.082)	0.139 (0.079)	0.142 (0.076)
GINSE		0.308*** (0.039)				0.231*** (0.041)	0.131** (0.041)		
GINSE ²		−0.121*** (0.029)				−0.083** (0.030)	−0.042 (0.028)		
GINRE			0.329*** (0.040)					0.328*** (0.041)	0.241*** (0.041)
GINRE ²			−0.129*** (0.026)					−0.104*** (0.026)	−0.072*** (0.026)
GDC					0.397*** (0.037)		0.325*** (0.035)		0.251*** (0.033)
Constant	−0.125 (0.173)	0.058 (0.170)	−0.031 (0.166)	−0.348 (0.308)	−0.298 (0.158)	−0.214 (0.156)	−0.229 (0.154)	−0.253 (0.147)	−0.246 (0.146)
R ²	0.059	0.219	0.258	0.078	0.149	0.195	0.168	0.248	0.304
F	5.982***	19.721***	24.111***	7.892***	13.623***	20.422***	13.810***	23.351***	26.542***
ΔR ²		0.161	0.199		0.149	0.085	0.078	0.176	0.040
ΔF		58.230***	75.127***		108.373***	29.996***	64.052**	59.470***	36.214***
VIF	1.020 ≤ VIF ≤ 2.190								

Note: Standardized regression coefficients are reported. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (two-tailed). GINSE = Green Innovation Network Structural Embeddedness; GINRE = Green Innovation Network Relational Embeddedness; GDC = Green Dynamic Capabilities; ESGP = ESG Performance; OA = Organizational Agility.

embeddedness corresponds to a 6.5 % raw score increase, while the near-zero effect of 0.024 at high embeddedness translates into only a 1.5 % improvement. This 87 % reduction in the mediating effect (from 0.187 to 0.024) suggests that firms with excessively high structural embeddedness gain minimal ESG benefits from their green capability development. Contextually, our sample data show that approximately 26 % of firms ($n = 147$) fall above the structural embeddedness turning point (standardized value > 1.488), indicating a non-negligible portion of enterprises may be overinvesting in network positioning without commensurate performance returns. In contrast, the relational embeddedness pathway exhibits a more gradual decline in mediation (from 0.153 to 0.019, a 88 % reduction), but with the moderate-level effect (0.085) remaining practically meaningful—equivalent to a 5.5 % raw ESG score increase—suggesting that relational quality sustains capability-based ESG gains more effectively than structural prominence. The near-complete loss of mediation at high structural embeddedness

can be attributed to two interrelated mechanisms: absorptive saturation and governance overload. At high embeddedness, firms accumulate redundant green resources that exceed their absorption capacity, disrupting the conversion of external inputs into internal capabilities (Cohen and Levinthal, 1990). Additionally, administrative complexity from coordinating with numerous network partners creates inertia, slowing down the iterative learning required for dynamic capability development. For example, several large SOEs in our sample reported “coordination fatigue” from participating in over 15 green innovation alliances, leading to delayed implementation of capability-building initiatives despite abundant network resources. This aligns with innovation ecosystem research highlighting that excessive network density can stifle value creation by increasing transaction costs and diluting strategic focus (Adner, 2017).

Notably, the turning points identified (standardized values 1.286–1.555) fall within the observed sample range, confirming the

Table 7
Quadratic functional forms, slope functions, and turning point analysis.

Path	Quadratic Function	Slope Function	Turning Point
1. GINSE → GDC	$GDC = -0.124GINSE^2 + 0.319GINSE + 0.058$	$S = -0.248GINSE + 0.319$	$GINSE = 1.286$
2. GINSE → ESGP	$ESGP = -0.082GINSE^2 + 0.244GINSE - 0.231$	$S = -0.164GINSE + 0.244$	$GINSE = 1.488$
3. GINRE → GDC	$GDC = -0.128GINRE^2 + 0.336GINRE - 0.025$	$S = -0.256GINRE + 0.336$	$GINRE = 1.313$
4. GINRE → ESGP	$ESGP = -0.109GINRE^2 + 0.339GINRE - 0.269$	$S = -0.218GINRE + 0.339$	$GINRE = 1.555$

Note: Each path follows a quadratic form confirming an inverted U-shape. First derivatives (S) become zero at specific turning points, all of which fall within the observed standardized value ranges, validating the nonlinear hypotheses. GINSE = Green Innovation Network Structural Embeddedness; GINRE = Green Innovation Network Relational Embeddedness; GDC = Green Dynamic Capabilities; ESGP = ESG Performance; OA = Organizational Agility.

theoretical existence of nonlinear inflection points. While this suggests many firms in the sample are still in the rising phase of the inverted U-curve, these turning points serve as critical early warnings: as ESG regulations tighten and collaborative intensity grows, firms approaching these thresholds may face increasing risk of diminishing returns. Practically, some high-embeddedness firms in our sample—particularly large state-owned enterprises (SOEs) participating in national ESG pilot programs—already exhibit signs of performance plateau or coordination overload (e.g., redundant green project investments, conflicting ESG disclosure standards), lending empirical credibility to the “curse” narrative even if not yet widespread. These results support Hypotheses H4a and H4b, demonstrating that green dynamic capabilities play a nonlinear mediating role in the relationship between network embeddedness and ESG performance.

4.6. Test of moderation and moderated mediation effects

To illustrate the moderating role of organizational agility, we followed the procedure recommended by Sun et al. (2018). Specifically, we examined the significance of the interaction terms between the squared

term of embeddedness and organizational agility in hierarchical regression models. The general regression model was specified as: $Y = \beta_0 + \beta_1X + \beta_2X^2 + \beta_3XZ + \beta_4X^2Z + \beta_5Z$, where Z represents organizational agility, and XZ and X^2Z represent the interactions between organizational agility and the linear and quadratic terms of embeddedness, respectively. A significant β_4 indicates a moderation effect on the curvature of the U-shape, with a negative β_4 suggesting a steeper curve, whereas a positive β_4 indicating a flatter one. As shown in Table 8, in Model 10 the interaction term between structural embeddedness (squared) and organizational agility was not significant ($\beta = -0.041$, $p > 0.05$), indicating that Hypothesis H5a was not supported. In Model 11, the interaction between relational embeddedness (squared) and organizational agility was significant ($\beta = -0.073$, $p < 0.05$). At high levels of agility (+1 SD), the curvature coefficient was -0.196 , while at low agility (−1 SD) it was -0.041 , supporting Hypothesis H5b. In Model 12, the interaction between structural embeddedness (squared) and agility had no significant effect on green dynamic capabilities ($\beta = -0.029$, $p > 0.05$), providing no support for Hypothesis H6a. However, in Model 13, the interaction between relational embeddedness (squared) and agility significantly affected green dynamic capabilities ($\beta = -0.091$, $p < 0.05$), and the curvature was more pronounced under high agility (-0.184) than under low agility (-0.026), supporting Hypothesis H6b. These results indicate that organizational agility significantly amplifies the inverted U-shaped effects in the relational embeddedness pathway, but has limited influence in the structural pathway.

To illustrate the moderating effect of organizational agility more intuitively, this study plotted the corresponding quadratic interaction graphs. Specifically, organizational agility was set at one standard deviation below the mean (M−1SD) to represent low agility and one standard deviation above the mean (M+1SD) to represent high agility. As shown in Figs. 2 and 3, the curvature is noticeably steeper under conditions of high organizational agility, providing further empirical support for Hypotheses H5b and H6b. To test moderated mediation, we followed the three-step procedure proposed by Cheung and Lau (2015). First, we tested whether the squared interaction terms significantly affected ESG performance. Second, we tested whether the same terms affected green dynamic capabilities. Third, we assessed whether green dynamic capabilities significantly predicted ESG performance when

Table 8
Hierarchical regression results for moderating and moderated mediation effects of organizational agility.

Variable	ESGP		GDC		ESGP	
	Model 10	Model 11	Model 12	Model 13	Model 14	Model 15
Size	0.183 (0.099)	0.078 (0.096)	0.128 (0.094)	0.037 (0.094)	0.141 (0.096)	0.064 (0.093)
Age	0.043 (0.049)	0.089* (0.045)	0.019 (0.051)	0.073 (0.050)	0.041 (0.048)	0.097 (0.045)
SOE	−0.312** (0.109)	−0.264** (0.097)	−0.266* (0.112)	−0.199 (0.108)	−0.239* (0.099)	−0.213* (0.098)
PE	0.042 (0.107)	0.095 (0.099)	−0.114 (0.102)	−0.036 (0.100)	0.046 (0.105)	0.039 (0.095)
IT	0.109 (0.072)	0.089 (0.069)	0.271*** (0.073)	0.243*** (0.072)	0.019 (0.069)	0.017 (0.067)
ICP	0.043 (0.075)	−0.096 (0.070)	−0.095 (0.072)	−0.032 (0.078)	0.118 (0.074)	0.112 (0.071)
GINSE	0.192*** (0.038)		0.302*** (0.041)		0.085* (0.041)	
GINSE ²	−0.108*** (0.027)		−0.131*** (0.027)		−0.077* (0.037)	
GINRE		0.221*** (0.039)		0.306*** (0.042)		0.153*** (0.039)
GINRE ²		−0.117*** (0.026)		−0.096** (0.027)		−0.087** (0.018)
OA	0.312*** (0.046)	0.328*** (0.043)	0.051 (0.046)	0.089 (0.047)	0.301** (0.050)	0.317*** (0.042)
GINSE × OA	0.147*** (0.036)		0.063 (0.038)		0.132** (0.036)	
GINSE ² × OA	−0.041 (0.028)		−0.029 (0.026)		−0.032 (0.028)	
GINRE × OA		0.151*** (0.041)		0.088* (0.042)		0.129*** (0.041)
GINRE ² × OA		−0.073* (0.024)		−0.091* (0.026)		−0.055* (0.023)
GDC					0.306** (0.055)	0.229*** (0.041)
Constant	−0.109 (0.149)	−0.210 (0.142)	0.063 (0.158)	−0.096 (0.151)	−0.126 (0.143)	−0.186 (0.140)
R ²	0.268	0.362	0.232	0.282	0.343	0.398
F	18.201***	28.931***	15.181***	19.476***	23.744***	31.521***
ΔR ²	0.092	0.108	0.018	0.047	0.082	0.046
ΔF	25.431***	30.192***	2.208	7.985**	63.556***	32.211***
VIF	1.032 ≤ VIF ≤ 2.201					

Note: Unstandardized coefficients reported; standard errors in parentheses. GINSE = Green Innovation Network Structural Embeddedness; GINRE = Green Innovation Network Relational Embeddedness; GDC = Green Dynamic Capabilities; ESGP = ESG Performance; OA = Organizational Agility. ***p < 0.001, **p < 0.01, *p < 0.05 (two-tailed).

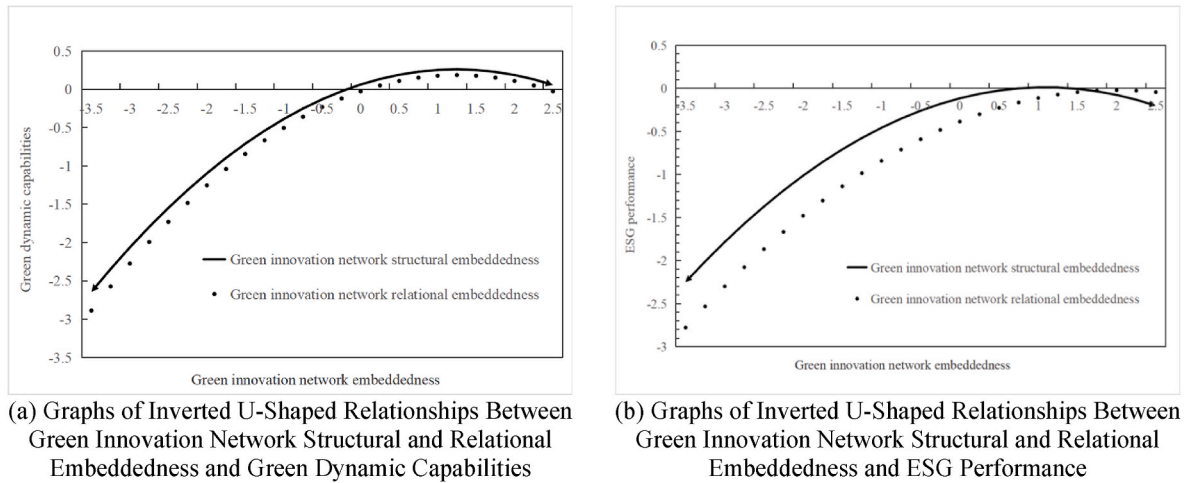


Fig. 2. Graphs of inverted U-shaped direct relationships.

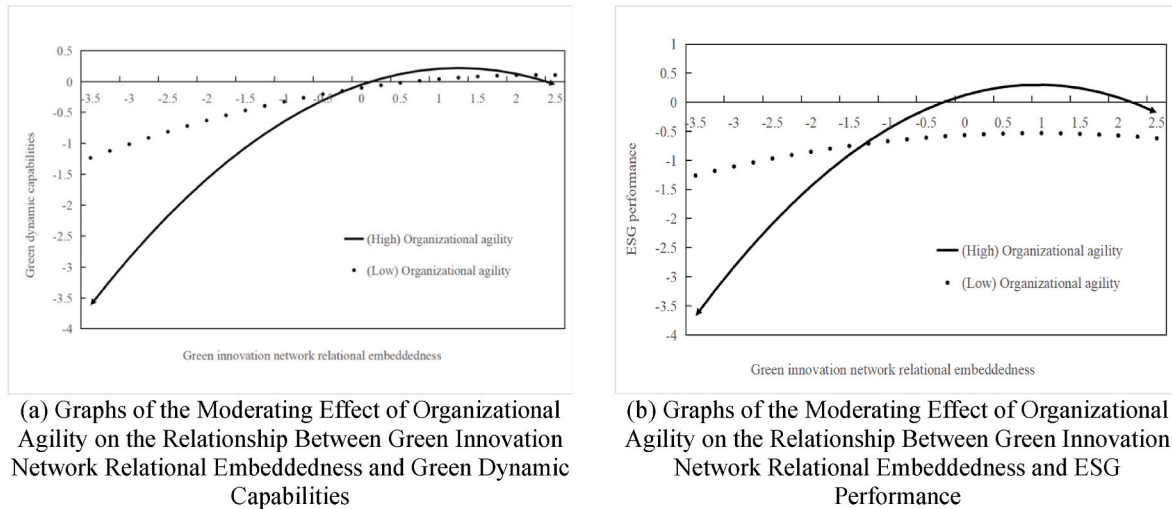


Fig. 3. Graphs of the moderating effects of organizational agility.

controlling for interaction terms. As shown in Table 7, Models 10, 12, and 14 indicated that the interaction terms related to structural embeddedness were not significant ($\beta = -0.041$, $p > 0.05$), thus failing to support Hypothesis H7a. In contrast, Models 11, 13, and 15 showed significant interaction effects in both ESG performance ($\beta = -0.073$, $p < 0.05$) and green dynamic capabilities ($\beta = -0.091$, $p < 0.05$), with green dynamic capabilities also significantly predicting ESG performance ($\beta = 0.229$, $p < 0.001$), satisfying all three criteria. Therefore, Hypothesis H7b was supported, confirming that organizational agility moderates the relationship between relational embeddedness and ESG performance through the mediation of green dynamic capabilities.

4.7. Robustness checks

To address potential concerns regarding sample heterogeneity and selection bias, we conducted robustness checks using two methods. First, we performed a subgroup analysis across firm size, ownership type, and industry category. We split the sample by firm size (large enterprises vs. SMEs) and re-estimated the core models. The inverted U-shaped relationships between network embeddedness (structural and relational) and ESG performance remained significant in both subgroups (large enterprises: structural embeddedness squared term $\beta = -0.079$, $p < 0.01$; relational embeddedness squared term $\beta = -0.098$, $p < 0.001$; SMEs: structural embeddedness squared term $\beta = -0.085$, $p < 0.01$;

relational embeddedness squared term $\beta = -0.102$, $p < 0.001$). The mediating role of green dynamic capabilities was also confirmed in both groups, with no significant differences in effect sizes. Second, we compared state-owned enterprises (SOEs) and private enterprises. The key nonlinear relationships and mediation effects held in both subgroups, though the turning point for relational embeddedness was slightly higher in SOEs (1.621) than in private firms (1.489), possibly due to SOEs' stronger institutional support for network collaboration. Third, we separated firms into strategic emerging energy industries (e.g., wind, solar, energy storage) and traditional energy industries (e.g., coal, oil). The inverted U-shaped relationships remained significant in both groups, but the negative effects of excessive embeddedness were more pronounced in traditional industries (structural embeddedness squared term $\beta = -0.112$, $p < 0.001$) than in emerging sectors ($\beta = -0.076$, $p < 0.01$), suggesting that traditional firms face greater coordination costs in over-embedded networks. These robustness checks confirm that the core theoretical relationships are stable across different subgroups, mitigating concerns about sample composition bias.

Second, we conducted robustness checks using Response Surface Analysis (RSA) to further validate the nonlinearity of the relationships. The RSA plots visually confirm the inverted U-shaped effects between structural and relational embeddedness and both green dynamic capabilities and ESG performance. Fig. 4(a) and (b) show the effects of network embeddedness on green dynamic capabilities and ESG

performance, respectively, with the optimal levels of embeddedness clearly highlighted. Fig. 4(c) and (d) illustrate the moderating effect of organizational agility on the relationship between relational embeddedness and both green dynamic capabilities and ESG performance, further reinforcing the inverted U-shape. These additional analyses ensure the stability of our results and mitigate concerns regarding potential bias from sample composition.

5. Discussions

5.1. Theoretical contributions

This study makes several theoretical contributions to the literature on green innovation, dynamic capabilities, and organizational agility, particularly in the context of energy enterprises undergoing a green transformation.

First, we extend the network embeddedness theory by demonstrating that structural and relational embeddedness in green innovation networks have distinct nonlinear effects on ESG performance. Whereas previous studies have predominantly assumed a linear relationship between network embeddedness and organizational outcomes (Ahuja, 2000; Pitelis et al., 2024), our findings highlight that both structural and relational embeddedness exhibit inverted U-shaped patterns in their influence on ESG performance (H1a, H1b). Specifically, excessive structural embeddedness leads to coordination overload and resource redundancy, whereas over-reliance on relational embeddedness may cause normative conformity and reduced strategic flexibility. This insight provides a deeper understanding of how network density and partner interdependence can positively or negatively influence firms' ability to adapt to green innovation demands and achieve sustainable performance. By introducing this nonlinearity, we offer a more nuanced framework for managing green networks in industries facing

sustainability pressures.

Second, we contribute to the dynamic capabilities literature by integrating green dynamic capabilities (GDC) as a key mediating mechanism between green innovation network embeddedness and ESG performance (H2a, H2b, H4a, H4b). While the literature on dynamic capabilities often focuses on general resource reconfiguration (Pitelis et al., 2024), our study emphasizes the specificity of green capabilities in managing environmental risks, seizing eco-innovation opportunities, and reconfiguring internal processes to align with sustainability goals. We show that green dynamic capabilities are most effectively developed at moderate levels of network embeddedness and play a central role in translating network resources into tangible ESG outcomes. This extension deepens our understanding of how firms develop specialized sustainability-oriented dynamic capabilities and how these capabilities influence long-term environmental and social performance (Helfat and Peteraf, 2003; Wu et al., 2025).

Third, we show that organizational agility significantly amplifies the relationship between relational embeddedness and both green dynamic capabilities and ESG performance (H5b, H6b) but does not affect the structural embeddedness paths in the same way (H5a, H6a). This finding suggests that agility works best in environments characterized by high relational trust and interaction, where firms can quickly adapt and respond to external environmental challenges (Gidage and Bhidé, 2025; Wang and Zhang, 2025b). By contrast, structural embeddedness, which is more rigid and resource-bound, reduces the effectiveness of organizational agility. This contribution advances the literature by showing that organizational agility is context-dependent, enhancing performance when embeddedness fosters knowledge exchange and adaptive collaboration, but not when it is too formalized or top-down (Xie et al., 2024b; Zhou et al., 2020).

Fourth, this study contributes to the understanding of green innovation networks and sustainability performance by highlighting the

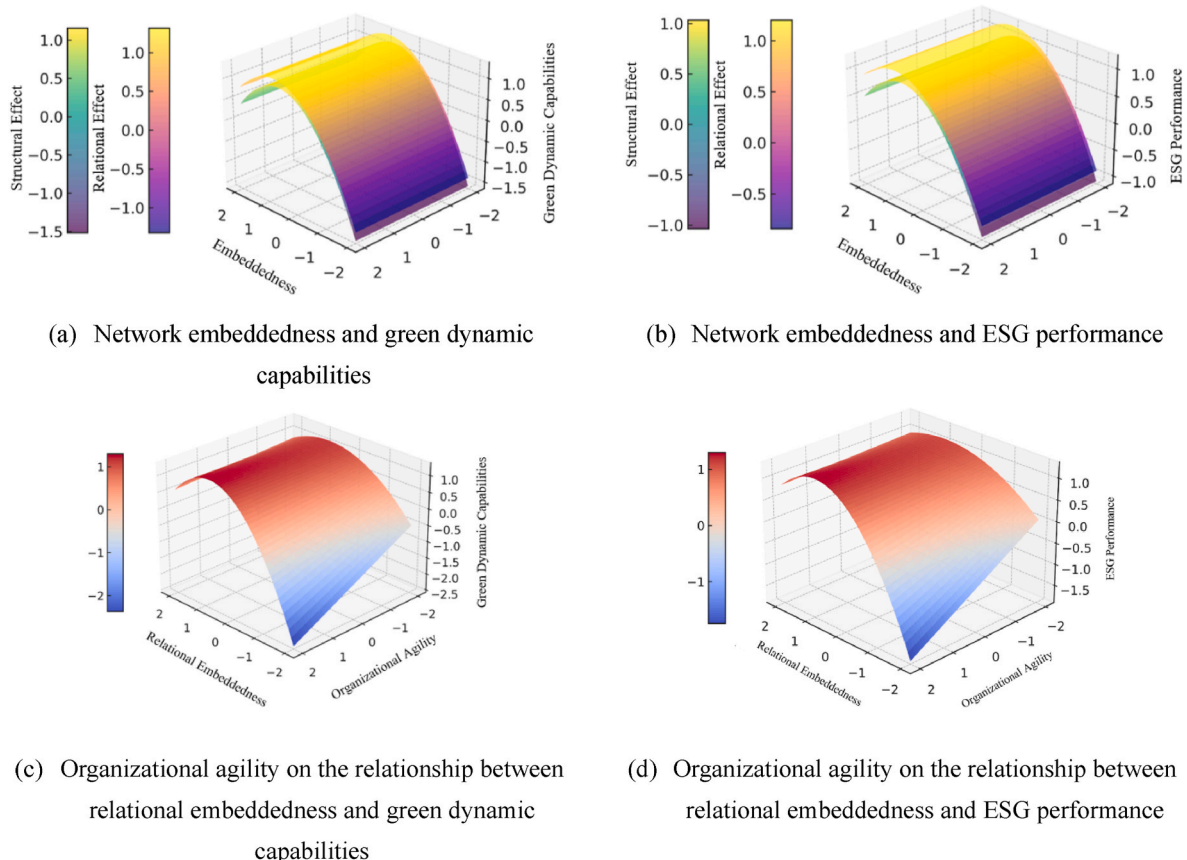


Fig. 4. Graphs of the response surface analysis.

asymmetric effects of organizational capabilities and network types. The distinct roles of structural and relational embeddedness, the mediating role of green dynamic capabilities, and the moderating role of organizational agility are all revealed to play specific, context-dependent roles in enhancing or constraining green innovation outcomes. This nuanced insight into network dynamics and capability development adds significant value to the current literature, especially by providing a practical guide for managers on how to configure green networks and build dynamic capabilities to achieve ESG goals.

In summary, this study expands the theoretical understanding of network embeddedness and dynamic capabilities by introducing nonlinear and context-specific pathways for green innovation networks. Our findings offer new insights into the complex and interdependent relationships between network characteristics, capability development, and performance outcomes, particularly in the energy sector. This research not only enriches the theoretical dialogue in sustainability and innovation management but also provides a comprehensive framework for understanding how firms in green transition contexts can effectively manage their innovation networks and dynamic capabilities to achieve superior ESG performance.

5.2. Practical implications

This study focuses on the network behavior and capability-building mechanisms of energy enterprises in the context of green transition, proposing and validating novel hypotheses regarding the relationship mechanisms among green innovation network embeddedness, green dynamic capabilities, and ESG performance. It emphasizes nonlinear pathways and dynamic capability processes, contributing five key theoretical advancements to the literature on green management, organizational theory, and sustainable development.

First, we propose and validate a theoretical framework demonstrating an inverted U-shaped nonlinear relationship between green innovation network embeddedness and ESG performance (H1a, H1b). For the first time in the green management field, this study distinguishes the differential impact pathways of structural and relational embeddedness, enriching the theoretical understanding of how green networks influence organizational performance. Unlike prior studies that predominantly focus on the linear effects of network embeddedness (Yin and Zhao, 2024), we find that both structural and relational embeddedness exhibit diminishing marginal returns beyond a certain threshold, potentially suppressing ESG performance. Excessive structural embeddedness may lead to information redundancy, control rigidity, and coordination burdens, while overly close relational embeddedness may foster inertial trust and a lack of heterogeneity. This finding provides a theoretical basis for identifying the "optimal embeddedness range" in green network management and offers a logical foundation for explaining the phenomenon of green collaboration failures.

Second, this study reveals an inverted U-shaped mechanism through which green innovation network embeddedness impacts the development of green dynamic capabilities (H2a, H2b), extending the antecedents of dynamic capability theory in green contexts. We find that moderate levels of both structural and relational embeddedness most effectively facilitate the development of green dynamic capabilities, while high levels significantly inhibit it. This finding challenges traditional assumptions that view network embeddedness as a consistently positive resource source, demonstrating that capability building is constrained by network "absorption saturation" and "path dependence." This expands the boundary conditions of the "resource orchestration–capability evolution" pathway.

Third, this study validates the nonlinear mediating mechanism of green dynamic capabilities in the relationship between network embeddedness and ESG performance (H3, H4a, H4b), positioning green capabilities as a critical mediating bridge for network value transformation and enriching the mechanistic explanation of green strategy implementation. We find that green dynamic capabilities significantly

and positively influence ESG performance (H3), with the strongest mediating effect at moderate embeddedness levels, nearly disappearing at high embeddedness levels. This suggests that network embeddedness alone is insufficient to enhance performance; the key lies in whether firms possess the mechanistic foundation to internalize external green resources into organizational capabilities, deepening the theoretical significance of capability transformation logic in green governance.

Fourth, this study identifies significant differential moderating effects of organizational agility in the pathways of structural and relational embeddedness (H5a–H6b), advancing the mechanistic application of agility theory in green management. In the relational embeddedness pathway, agility significantly enhances the marginal positive effects on ESG performance and green capabilities (H5b, H6b supported), whereas its moderating role in the structural embeddedness pathway is insignificant (H5a, H6a unsupported). This heterogeneity indicates that agility is more readily activated in highly interactive, trust-based relational networks but has limited effects in structurally rigid and control-heavy networks, enriching the explanatory logic of the "organizational context–capability mobilization–strategic performance" moderation chain.

Finally, through systematic integration of the above mechanisms, we construct a nonlinear interaction framework of "network embeddedness–capability building–performance outcomes" and empirically validate it in the green energy context. The theoretical contributions closely align with the hypotheses: H1–H2 correspond to the inverted U-shaped network pathways, H3–H4 reveal the nonlinear mediating mechanisms, and H5–H6 reflect differentiated moderating effects. Overall, this study not only deepens the mechanistic foundation of green innovation network research but also advances the theoretical integration of dynamic capabilities and organizational agility in green transition scenarios, providing significant insights for the development of green management theory and the optimization of green strategic practices.

6. Conclusion

Drawing on a sample of energy enterprises in Northwestern China and grounded in resource orchestration theory and the dynamic capability perspective, this study systematically investigates the mechanisms through which green innovation network embeddedness influences ESG performance. By constructing a theoretical framework incorporating the mediating role of green dynamic capabilities and the moderating role of organizational agility, and applying hierarchical regression and nonlinear mediation analyses, this study empirically validates the nonlinear effects and complex transmission pathways of network embeddedness, offering valuable insights for both theoretical research and practical application in green transformation.

Empirical findings confirm that both structural and relational dimensions of green innovation network embeddedness exhibit inverted U-shaped relationships with green dynamic capabilities and ESG performance, supporting Hypotheses H1a, H1b, H2a, and H2b. Moderate structural embeddedness enhances access to diverse green technologies and institutional resources by fostering connections with green technology providers, research institutions, and policy platforms, thereby facilitating capability development and ESG improvement (Gulati and Sytch, 2008). However, excessive embeddedness leads to resource redundancy, technological fragmentation, and increased coordination costs, ultimately weakening integration efficiency and suppressing ESG performance. Relational embeddedness, at moderate levels, promotes deep trust-based collaboration and knowledge sharing, which supports the internalization of green governance logics and enhances both green capabilities and ESG performance (Huang et al., 2025). When over-embedded, however, cognitive lock-in and strategic drift emerge, impeding capability formation and lowering performance. Moreover, green dynamic capabilities are found to mediate the relationship between network embeddedness and ESG performance in a nonlinear fashion, confirming Hypotheses H4a and H4b. At moderate levels of

embeddedness, green dynamic capabilities effectively convert network resources into performance outcomes, whereas at high levels, capability redundancy and resource overload diminish the mediating effect. Regarding the moderating role of organizational agility, results show that agility significantly enhances the curvature of the inverted U-shaped relationships in the relational embeddedness pathway, supporting Hypotheses H5b, H6b, and H7b, while no significant moderation was observed in the structural pathway, and Hypotheses H5a, H6a, and H7a were not supported. These findings reveal the nonlinear and contingency-based nature of green innovation network embeddedness and offer strategic insights for optimizing green transformation practices.

7. Limitations and future research

This study develops a theoretical framework and conducts empirical tests to examine the impact pathways of green innovation network embeddedness on ESG performance in energy enterprises. While the findings offer robust insights, several limitations remain and warrant further exploration. First, this study focuses primarily on how green innovation network embeddedness affects ESG performance, without systematically investigating its broader implications for other dimensions of firm management. For instance, such embeddedness may also influence green technological innovation performance, the evolution of green business models, or the efficiency of green strategy implementation. Moreover, beyond the energy sector, other high carbon-intensive or ecologically dependent industries such as agriculture, manufacturing, transportation, and services face similar challenges during green transitions. Future research could extend the scope by exploring how green innovation network embeddedness affects various innovation performance dimensions across different industries, thereby enriching the theoretical landscape of green management and deepening the understanding of how such embeddedness operates under diverse organizational conditions.

Second, this study relies on cross-sectional survey data for empirical analysis. Although this approach allows for the identification of static associations among variables, it severely limits causal inference—particularly for the mediation model examining green dynamic capabilities and ESG performance outcomes. Green dynamic capabilities, by nature, emerge through iterative, path-dependent processes involving repeated cycles of partner interaction, knowledge absorption, and organizational adaptation. Their development unfolds over time, shaped by ongoing collaboration and adaptive learning within green networks. Likewise, ESG performance improvements—especially in dimensions such as carbon reduction, governance reform, or stakeholder relationship management—often exhibit time lags relative to capability upgrades. For example, investments in green dynamic capabilities may take 1–3 years to translate into measurable ESG gains. Our cross-sectional design captures only a snapshot of these ongoing processes, meaning the observed relationships—while statistically robust—should be interpreted as associative rather than processual or time-ordered. We cannot definitively confirm the sequential causality implied in our theoretical model nor account for potential reverse causality. Additionally, the nonlinear effects identified may reflect different stages of capability development across firms rather than stable patterns, which a single-timepoint dataset cannot disentangle. To address this limitation, future research should adopt longitudinal or panel data designs, tracking firms over 3–5 years to map the co-evolution of network embeddedness, green dynamic capabilities, and ESG performance. Such designs could capture: (1) how changes in network structure precede shifts in capability development; (2) lagged effects of capability upgrades on ESG outcomes; and (3) feedback loops. Combining survey data with archival sources—such as third-party ESG ratings, carbon emission reports, or digital trace data from green collaboration platforms—would further strengthen temporal precision and reduce reliance on self-reported measures.

Third, the sample composition raises concerns about generalizability, particularly regarding representation of traditional energy sub-sectors. Our sample overrepresents firms in strategic emerging energy industries and underrepresents traditional coal and oil enterprises, which face the most acute green transition challenges. This tilt may stem from our collaboration with green technology service providers and industry associations, which tend to engage more with progressive or reform-oriented firms. As a result, our findings may underestimate the variance in network embeddedness and performance gaps among resource-constrained or conservative traditional energy firms, where the “curse” effects of over-embeddedness could be more severe due to weaker absorptive capacity. Future research should adopt quota sampling or probability-based approaches, incorporating national-level energy enterprise registries, to ensure proportional representation of sub-sectors and transformation stages. This would enhance the generalizability of findings across China’s diverse energy landscape.

Fourth, this research is based on data collected from energy enterprises in Northwest China. Under the guidance of the national dual-carbon strategy, these firms are simultaneously undergoing industrialization, digital transformation, and low-carbon transition. Their network structures and institutional embeddedness processes exhibit distinct Chinese characteristics and transitional features. Therefore, the generalizability of this study’s findings to firms operating in other countries or institutional settings remains to be verified. Future research could expand the data sources to include enterprises embedded in varied institutional arrangements and green regulatory environments, to examine the contextual adaptability and effectiveness of green innovation network embeddedness. Additionally, future work may introduce alternative moderating or mediating variables to further enrich the theoretical explanation of how such embeddedness contributes to sustainable competitive advantage, and to advance the cross-contextual extension and theoretical evolution of green management research.

CRedit authorship contribution statement

Hao Dong: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Liting Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Xun Wu:** Writing – original draft, Software, Methodology, Data curation. **Ke Hou:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization.

Declaration

Authors declare no competing interests.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Adner, R., 2017. Ecosystem as structure: an actionable construct for strategy. *J. Manag.* 43, 39–58. <https://doi.org/10.1177/0149206316678451>.
- Ahuja, G., 2000. The duality of collaboration: inducements and opportunities in the formation of interfirm linkages. *Strateg. Manag. J.* 21, 317–343. [https://doi.org/10.1002/\(sici\)1097-0266\(200003\)21:3<317::aid-smj90>3.0.co;2-b](https://doi.org/10.1002/(sici)1097-0266(200003)21:3<317::aid-smj90>3.0.co;2-b).
- Aragón-Correa, J.A., Marcus, A.A., Vogel, D., 2020. The effects of mandatory and voluntary regulatory pressures on firms' environmental strategies: a review and recommendations for future research. *Acad. Manag. Ann.* <https://doi.org/10.5465/annals.2018.0014>.
- Baah, C., Jin, Z., Tang, L., 2020. Organizational and regulatory stakeholder pressures friends or foes to green logistics practices and financial performance: investigating corporate reputation as a missing link. *J. Clean. Prod.* 247, 119125. <https://doi.org/10.1016/j.jclepro.2019.119125>.
- Barbieri, N., Marzocchi, A., Rizzo, U., 2020. Knowledge sources and impacts on subsequent inventions: do green technologies differ from non-green ones? *Res. Pol.* 49, 103901. <https://doi.org/10.1016/j.respol.2019.103901>.
- Barney, J.B., 2018. Why resource-based theory's model of profit appropriation must incorporate a stakeholder perspective. *Strateg. Manag. J.* 39, 3305–3325. <https://doi.org/10.1002/smj.2949>.
- Caldera, H.T.S., Desha, C., Dawes, L., 2019. Evaluating the enablers and barriers for successful implementation of sustainable business practice in 'lean' SMEs. *J. Clean. Prod.* 218, 575–590. <https://doi.org/10.1016/j.jclepro.2019.01.239>.
- Carnes, C.M., Chirico, F., Hitt, M.A., Huh, D.W., Pisano, V., 2017. Resource orchestration for innovation: structuring and bundling resources in Growth- and maturity-stage firms. *Long. Range Plan.* 50, 472–486. <https://doi.org/10.1016/j.lrp.2016.07.003>.
- Cavusgil, S.T., Deligonul, S.Z., 2024. Dynamic capabilities framework and its transformative contributions. *J. Int. Bus. Stud.* 56, 33–42. <https://doi.org/10.1057/s41267-024-00758-8>.
- Chen, M.-J., Michel, J.G., Lin, W., 2021. Worlds apart? Connecting competitive dynamics and the resource-based view of the firm. *J. Manag.* 47, 1820–1840. <https://doi.org/10.1177/01492063211000422>.
- Cheung, G.W., Lau, R.S., 2015. Accuracy of parameter estimates and confidence intervals in moderated mediation models: a comparison of regression and latent moderated structural equations. *Organ. Res. Methods* 20, 746–769. <https://doi.org/10.1177/1094428115595869>.
- Cohen, W.M., Levinthal, D.A., 1990. Absorptive capacity: a new perspective on learning and innovation. *Adm. Sci. Q.* 35, 128–152. <https://doi.org/10.2307/2393553>.
- Ding, C.G., Chen, C.-F., Jane, T.-D., 2023. Improving the performance of the unmeasured latent method construct technique in common method variance detection and correction. *J. Organ. Behav.* 44, 519–542. <https://doi.org/10.1002/job.2673>.
- D'Oria, L., Crook, T.R., Ketchen Jr, D.J., Sirmon, D.G., Wright, M., 2021. The evolution of resource-based inquiry: a review and meta-analytic integration of the strategic resources-actions-performance pathway. *J. Manag.* 47, 1383–1429. <https://doi.org/10.1177/0149206321994182>.
- Fernando, Y., Chiappetta Jabbour, C.J., Wah, W.-X., 2019. Pursuing green growth in technology firms through the connections between environmental innovation and sustainable business performance: does service capability matter? *Resour. Conserv. Recycl.* 141, 8–20. <https://doi.org/10.1016/j.resconrec.2018.09.031>.
- Gao, B., Kong, X., Liu, G., Xiang, T., Gao, Y., Luo, S., Wang, Y., 2024. Monitoring high-carbon industry enterprise emission in carbon market: a multi-trusted approach using externally available big data. *J. Clean. Prod.* 466, 142729. <https://doi.org/10.1016/j.jclepro.2024.142729>.
- Gidage, M., Bhide, S., 2025. ESG performance and green innovation nexus: insights from Indian manufacturing titans. *Bus. Strat. Environ.* <https://doi.org/10.1002/bse.70041>.
- Gidage, M., Gyamfi, B.A., Asongu, S.A., 2025. Environmental reporting and financial performance: evidence from the banking sector in BRICS countries. *Business Strategy Environ.* n/a. <https://doi.org/10.1002/bse.4358>.
- Gordano, S., Bauer, F., King, D.R., 2024. Environmental, social, and governance (ESG) acquisitions: deal structure and outcome. *Bus. Strat. Environ.* 33, 6010–6028. <https://doi.org/10.1002/bse.3791>.
- Granovetter, M., 1985. Economic action and social structure: the problem of embeddedness. *Am. J. Sociol.* 91, 481–510. <https://doi.org/10.1086/228311>.
- Gulati, R., Sytch, M., 2008. Does familiarity breed trust? Revisiting the antecedents of trust. *Manag. Decis. Econ.* 29, 165–190. <https://doi.org/10.1002/mde.1396>.
- Guo, R., Yin, H., Liu, X., 2023. Coopetition, organizational agility, and innovation performance in digital new ventures. *Ind. Mark. Manag.* 111, 143–157. <https://doi.org/10.1016/j.indmarman.2023.04.003>.
- Haans, R.F.J., Pieters, C., He, Z.-L., 2016. Thinking about U: theorizing and testing U- and inverted U-shaped relationships in strategy research. *Strateg. Manag. J.* 37, 1177–1195. <https://doi.org/10.1002/smj.2399>.
- Hair, J.F., Risher, J.J., Sarstedt, M., Ringle, C.M., 2019. When to use and how to report the results of PLS-SEM. *Eur. Bus. Rev.* 31, 2–24. <https://doi.org/10.1108/ebv-11-2018-0203>.
- Hayes, A.F., Preacher, K.J., 2010. Quantifying and testing indirect effects in simple mediation models when the constituent paths are nonlinear. *Multivariate Behav. Res.* 45, 627–660. <https://doi.org/10.1080/00273171.2010.498290>.
- Helfat, C.E., Peteraf, M.A., 2003. The dynamic resource-based view: capability lifecycles. *Strateg. Manag. J.* 24, 997–1010. <https://doi.org/10.1002/smj.332>.
- Huang, L., Liu, H., Oghenerobor, E., Chen, C.-S., 2025. Collaborating digitalization and green supply chain to promote green development of manufacturing firms – an analysis from a configurational perspective. *Chin. Manag. Stud.* <https://doi.org/10.1108/CMS-08-2024-0570>.
- Huang, R., Zhu, Z., Ruan, R., Lou, X., 2024. Linking low-carbon practices with ESG performances: exploration evidence from the configurational perspective. *J. Clean. Prod.* 435, 140532. <https://doi.org/10.1016/j.jclepro.2023.140532>.
- Jiang, Y., Ni, H., Ni, Y., Guo, X., 2023. Assessing environmental, social, and governance performance and natural resource management policies in China's dual carbon era for a green economy. *Resour. Policy* 85, 104050. <https://doi.org/10.1016/j.resourpol.2023.104050>.
- Juo, W.-J., Wang, C.-H., 2022. Does green innovation mediate the relationship between green relational view and competitive advantage? *Bus. Strat. Environ.* 31, 2456–2468. <https://doi.org/10.1002/bse.3037>.
- Lampikoski, T., Westerlund, M., Rajala, R., Möller, K., 2014. Green innovation games: value-creation strategies for corporate sustainability. *Calif. Manag. Rev.* 57, 88–116. <https://doi.org/10.1525/cmr.2014.57.1.88>.
- Liao, Z., 2018. Institutional pressure, knowledge acquisition and a firm's environmental innovation. *Bus. Strat. Environ.* 27, 849–857. <https://doi.org/10.1002/bse.2036>.
- Liu, Y., Shao, X., Tang, M., Lan, H., 2021. Spatio-temporal evolution of green innovation network and its multidimensional proximity analysis: empirical evidence from China. *J. Clean. Prod.* 283, 124649. <https://doi.org/10.1016/j.jclepro.2020.124649>.
- Maksimov, V., Wang, S.L., Yan, S., 2019. Global connectedness and dynamic green capabilities in MNEs. *J. Int. Bus. Stud.* 53, 723–740. <https://doi.org/10.1057/s41267-019-00275-z>.
- Mao, H., Liu, S., Gong, Y., 2024. Balancing structural IT capabilities for organizational agility in digital transformation: a resource orchestration view. *IJOPM* 44, 315–344. <https://doi.org/10.1108/IJOPM-09-2022-0595>.
- Mariani, C., Caccialanza, A., Bugarcic, M., Slavkovic, M., Mancini, M., 2025. Enhancing project-based organization performance through ESG practices: the role of organizational agility. *Manag. Decis.* <https://doi.org/10.1108/MD-04-2024-0967> ahead-of-print.
- McFadyen, M.A., Cannella, A.A., 2004. Social capital and knowledge creation: diminishing returns of the number and strength of exchange relationships. *Australas. Mark. J.* 47, 735–746. <https://doi.org/10.5465/20159615>.
- Melander, L., Arvidsson, A., 2022. Green innovation networks: a research agenda. *J. Clean. Prod.* 357, 131926. <https://doi.org/10.1016/j.jclepro.2022.131926>.
- Mondal, S., Singh, S., Gupta, H., 2025. Dynamic capabilities and green strategy in green entrepreneurship and circular economy: a study. *Bus. Strat. Environ.* <https://doi.org/10.1002/bse.70030>.
- Moran, P., 2005. Structural vs. relational embeddedness: social capital and managerial performance. *Strateg. Manag. J.* 26, 1129–1151. <https://doi.org/10.1002/smj.486>.
- Omidvar, O., Safavi, M., Glaser, V.L., 2022. Algorithmic routines and dynamic inertia: how organizations avoid adapting to changes in the environment. *J. Manag. Stud.* 60, 313–345. <https://doi.org/10.1111/joms.12819>.
- Peng, H., Shen, N., Liao, H., Wang, Q., 2020a. Multiple network embedding, green knowledge integration and green supply chain performance—investigation based on agglomeration scenario. *J. Clean. Prod.* 259, 120821. <https://doi.org/10.1016/j.jclepro.2020.120821>.
- Peng, H., Shen, N., Liao, H., Wang, Q., 2020b. Multiple network embedding, green knowledge integration and green supply chain performance—investigation based on agglomeration scenario. *J. Clean. Prod.* 259, 120821. <https://doi.org/10.1016/j.jclepro.2020.120821>.
- Pitelis, C.N., Teece, D.J., Yang, H., 2024. Dynamic capabilities and MNE global strategy: a systematic literature review-based novel conceptual framework. *J. Manag. Stud.* 61, 3295–3326. <https://doi.org/10.1111/joms.13021>.
- Puriwat, W., Hoonsopon, D., 2021. Cultivating product innovation performance through creativity: the impact of organizational agility and flexibility under technological turbulence. *J. Manuf. Technol. Manag.* 33, 741–762. <https://doi.org/10.1108/jmtm-10-2020-0420>.
- Rabal-Conesa, J., Jiménez-Jiménez, D., Martínez-Costa, M., 2021. Organisational agility, environmental knowledge and green product success. *J. Knowl. Manag.* 26, 2440–2462. <https://doi.org/10.1108/JKM-06-2021-0486>.
- Ranganathan, R., Chen, J.S., Ghosh, A., 2025. Shaping ecosystem rules: complementarities, interdependencies, and firms' success in coordinating ecosystems via standard-setting. *Organ. Sci.* <https://doi.org/10.1287/orsc.2022.16136>.
- Sadovnikova, A., Pujari, A., 2017. The effect of green partnerships on firm value. *J. Acad. Market. Sci.* 45, 251–267. <https://doi.org/10.1007/s11747-016-0490-9>.
- Sahoo, S., Kumar, A., Upadhyay, A., 2023. How do green knowledge management and green technology innovation impact corporate environmental performance? Understanding the role of green knowledge acquisition. *Bus. Strat. Environ.* 32, 551–569. <https://doi.org/10.1002/bse.3160>.
- Salvato, C., Vassolo, R., 2017. The sources of dynamism in dynamic capabilities. *Strateg. Manag. J.* 39, 1728–1752. <https://doi.org/10.1002/smj.2703>.
- Sang, Y., Loganathan, K., Sukirthanandan, P., 2025. A study on the impact of corporate digital transformation on environmental, social, and governance (ESG) performance: mechanism analysis based on resource allocation efficiency and technological gap. *Sustainability* 17, 3308. <https://doi.org/10.3390/su17083308>.
- Shah, K.U., 2011. Organizational legitimacy and the strategic bridging ability of green alliances. *Bus. Strat. Environ.* 20, 498–511. <https://doi.org/10.1002/bse.706>.
- Simmering, M.J., Fuller, C.M., Richardson, H.A., Ocal, Y., Atinc, G.M., 2015. Marker variable choice, reporting, and interpretation in the detection of common method

- variance: a review and demonstration. *Organ. Res. Methods* 18, 473–511. <https://doi.org/10.1177/1094428114560023>.
- Singh, S.K., Giudice, M.D., Jabbour, C.J.C., Latan, H., Sohal, A.S., 2021. Stakeholder pressure, green innovation, and performance in small and medium-sized enterprises: the role of green dynamic capabilities. *Bus. Strat. Environ.* 31, 500–514. <https://doi.org/10.1002/bse.2906>.
- Sirmon, D.G., Hitt, M.A., Ireland, R.D., Gilbert, B.A., 2011. Resource orchestration to create competitive advantage: breadth, depth, and life cycle effects. *J. Manag.* 37, 1390–1412. <https://doi.org/10.1177/0149206310385695>.
- Sui, X., Hu, H., Wang, H., Hu, T., 2024. The impact of servitization transformation on the ESG performance of manufacturing firms. *Int. Rev. Econ. Finance* 96, 103582. <https://doi.org/10.1016/j.iref.2024.103582>.
- Sun, W., Yao, S., Govind, R., 2018. Reexamining corporate social responsibility and shareholder value: the Inverted-U-Shaped relationship and the moderation of marketing capability. *J. Bus. Ethics* 160, 1001–1017. <https://doi.org/10.1007/s10551-018-3854-x>.
- Tang, H., Zhao, R., Jiang, D., Chu, Z., 2025. Associate with the green to become greener: TMT network and corporate ESG performance. *Corp. Soc. Responsib. Environ. Manag.* <https://doi.org/10.1002/csr.3256>.
- Teece, D.J., 2007. Explicating dynamic capabilities: the nature and microfoundations of (sustainable) enterprise performance. *Strateg. Manag. J.* 28, 1319–1350. <https://doi.org/10.1002/smj.640>.
- Teeter, P., Sandberg, J., 2017. Constraining or enabling green capability development? How policy uncertainty affects organizational responses to flexible environmental regulations. *Br. J. Manag.* 28, 649–665. <https://doi.org/10.1111/1467-8551.12188>.
- Uddin, M., Rahman, A.A., 2012. Energy efficiency and low carbon enabler green IT framework for data centers considering green metrics. *Renew. Sustain. Energy Rev.* 16, 4078–4094. <https://doi.org/10.1016/j.rser.2012.03.014>.
- Ullah, S., Kukreti, M., Sami, A., Shaikat, M.R., 2024. Leveraging technological readiness and green dynamic capability to enhance sustainability performance in manufacturing firms. *J. Manuf. Technol. Manag.* 36, 714–730. <https://doi.org/10.1108/JMTM-05-2024-0268>.
- Wang, G.-J., Lin, X., Zhu, Y., Xie, C., Uddin, G.S., 2024. Corporate ESG performance and systemic risk: a network perspective. *Ann. Oper. Res.* <https://doi.org/10.1007/s10479-024-06163-6>.
- Wang, J., Xue, Y., Yang, J., 2023. Can proactive boundary-spanning search enhance green innovation? The mediating role of organizational resilience. *Bus. Strat. Environ.* 32, 1981–1995. <https://doi.org/10.1002/bse.3231>.
- Wang, S., Zhang, H., 2025a. Artificial intelligence digital employees and sustainable innovation in online retail: the mediating role of ambidextrous green innovation and the moderating role of ethical anxiety. *J. Retailing Consum. Serv.* 84, 104235. <https://doi.org/10.1016/j.jretconser.2025.104235>.
- Wang, S., Zhang, H., 2025b. Digital transformation and innovation performance in Small- and medium-sized enterprises: a systems perspective on the interplay of digital adoption, digital drive, and digital culture. *Systems* 13, 43. <https://doi.org/10.3390/systems13010043>.
- Wang, S., Zhang, H., 2024. Inter-organizational cooperation in digital green supply chains: a catalyst for eco-innovations and sustainable business practices. *J. Clean. Prod.* 472, 143383. <https://doi.org/10.1016/j.jclepro.2024.143383>.
- Woo, J., Fatima, R., Kibert, C.J., Newman, R.E., Tian, Y., Srinivasan, R.S., 2021. Applying blockchain technology for building energy performance measurement, reporting, and verification (MRV) and the carbon credit market: a review of the literature. *Build. Environ.* 205, 108199. <https://doi.org/10.1016/j.buildenv.2021.108199>.
- Wu, Z., Long, S., Wang, Y., 2025. ESG greenwashing and organizational resilience: exploring sustainable development paths from the perspective of unsystematic risk. *J. Environ. Manag.* 390, 126395. <https://doi.org/10.1016/j.jenvman.2025.126395>.
- Xia, D., Zhang, M., Yu, Q., Tu, Y., 2019. Developing a framework to identify barriers of green technology adoption for enterprises. *Resour. Conserv. Recycl.* 143, 99–110. <https://doi.org/10.1016/j.resconrec.2018.12.022>.
- Xie, X., Sun, J., Zhou, M., Yan, L., Chi, M., 2024a. Network embeddedness and manufacturing SMEs' green innovation performance: the moderating role of resource orchestration capability. *Bus. Process Manag. J.* 30, 884–908. <https://doi.org/10.1108/BPMJ-08-2023-0676>.
- Xie, X., Sun, J., Zhou, M., Yan, L., Chi, M., 2024b. Network embeddedness and manufacturing SMEs' green innovation performance: the moderating role of resource orchestration capability. *Bus. Process Manag. J.* 30, 884–908. <https://doi.org/10.1108/BPMJ-08-2023-0676>.
- Xu, B., 2024. Environmental regulations, technological innovation, and low carbon transformation: a case of the logistics industry in China. *J. Clean. Prod.* 439, 140710.1–140710.19.
- Yin, S., Zhao, Y., 2024. Digital green value co-creation behavior, digital green network embedding and digital green innovation performance: moderating effects of digital green network fragmentation. *Humanit. Soc. Sci. Commun.* 11, 228. <https://doi.org/10.1057/s41599-024-02691-5>.
- Yongsheng, X., Xiaole, Z., Wei, W., 2021. Coupling or lock-in? Co-evolution of cultural embeddedness and cluster innovation-exploratory case study of Shaoxing textile cluster. *Technol. Soc.* 67, 101765. <https://doi.org/10.1016/j.techsoc.2021.101765>.
- Yu, Z., Liu, Y., Yan, T., Zhang, M., 2024. Carbon emission efficiency in the age of digital economy: new insights on green technology progress and industrial structure distortion. *Bus. Strat. Environ.* 33, 4039–4057. <https://doi.org/10.1002/bse.3683>.
- Yuan, B., Cao, X., 2022. Do corporate social responsibility practices contribute to green innovation? The mediating role of green dynamic capability. *Technol. Soc.* 68, 101868. <https://doi.org/10.1016/j.techsoc.2022.101868>.
- Zhang, C., Farooq, U., Jamali, D., Alam, M.M., 2024. The role of ESG performance in the nexus between economic policy uncertainty and corporate investment. *Res. Int. Bus. Finance* 70, 102358. <https://doi.org/10.1016/j.ribaf.2024.102358>.
- Zhang, Q., Oo, B.L., Lim, B.T.H., 2022. Linking corporate social responsibility (CSR) practices and organizational performance in the construction industry: a resource collaboration network. *Resour. Conserv. Recycl.* 179, 106113. <https://doi.org/10.1016/j.resconrec.2021.106113>.
- Zhang, R., Liu, J., Cao, Z., 2023. Green innovation ecosystems: spatial organization mode and associated network renewal under coupling effect. *J. Clean. Prod.* 422, 138539. <https://doi.org/10.1016/j.jclepro.2023.138539>.
- Zhang, W., Ye, S., Mangla, S.K., Emrouznejad, A., Song, M., 2024. Smart platforming in automotive manufacturing for NetZero: intelligentization, green technology, and innovation dynamics. *Int. J. Prod. Econ.* 274, 109289. <https://doi.org/10.1016/j.ijpe.2024.109289>.
- Zhao, D., Wang, L., Gu, F.F., 2021. Reconsidering network embeddedness: effects on different forms of opportunism. *J. Bus. Res.* 131, 12–24. <https://doi.org/10.1016/j.jbusres.2021.03.029>.
- Zhao, S., Wang, Z., Arkorful, V.E., 2025. Political network embeddedness and ecological innovation: do absorptive capacity and dynamic capability have any influence? *Corp. Soc. Responsib. Environ. Manag.* 32, 2085–2103. <https://doi.org/10.1002/csr.3048>.
- Zhou, J., Lei, X., Yu, J., 2024. ESG rating divergence and corporate green innovation. *Bus. Strat. Environ.* 33, 2911–2930. <https://doi.org/10.1002/bse.3636>.
- Zhou, M., Govindan, K., Xie, X., 2020. How fairness perceptions, embeddedness, and knowledge sharing drive green innovation in sustainable supply chains: an equity theory and network perspective to achieve sustainable development goals. *J. Clean. Prod.* 260, 120950. <https://doi.org/10.1016/j.jclepro.2020.120950>.
- Zhu, L., Li, T., Wang, C., Huang, J., 2025. Corporate ESG performance and green innovation: moderating effect of shareholder activism. *J. Environ. Manag.* 383, 125413. <https://doi.org/10.1016/j.jenvman.2025.125413>.